



A critical review of critical power

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Abstract

The elegant concept of a hyperbolic relationship between power, velocity, or torque and time to exhaustion has rightfully captivated the imagination and inspired extensive research for over half a century. Theoretically, the relationship's asymptote along the time axis (critical power, velocity, or torque) indicates the exercise intensity that could be maintained for extended durations, or the “heavy–severe exercise boundary”. Much more than a critical mass of the extensive accumulated evidence, however, has persistently shown the determined intensity of critical power and its variants as being too high to maintain for extended periods. The extensive scientific research devoted to the topic has almost exclusively centered around its relationships with various endurance parameters and performances, as well as the identification of procedural problems and how to mitigate them. The prevalent underlying premise has been that the observed discrepancies are mainly due to experimental ‘noise’ and procedural inconsistencies. Consequently, little or no effort has been directed at other perspectives such as trying to elucidate physiological reasons that possibly underly and account for those discrepancies. This review, therefore, will attempt to offer a new such perspective and point out the discrepancies’ likely root causes.

Keywords Endurance · Fatigue · MLSS · MMSS · Critical torque · Critical velocity · Critical speed

Abbreviations

AWC	Anaerobic work capacity
CP	Critical power
CT	Critical torque
CS	Critical speed/velocity
CWR	Constant work rate (power)
D_{lim}	Distance limit for a given running duration
ICC	Intra-class correlation
La^-	Lactate
$[La^-]$	Lactate concentration
LMT	Lactate minimum test
MAP	Maximal aerobic power
MLSS	Maximal lactate steady state

MMSS	Maximal metabolic steady state
$P-t$	Power vs. time to exhaustion (relationship)
$P-1/t$	Power vs. 1/time relationship
RLT	Reverse lactate threshold test
rpm	Revolutions per minute
t_{lim}	Time limit (of a given power output, torque, or speed)
TTE	Time to exhaustion
VO_2	Oxygen consumption
VO_{2max}	Maximal oxygen consumption
$W-t$	Work vs. time relationship
W'	The curvature constant of the power–time curve/Maximal power output at a given endurance time point
W_{lim}	Work limit for a given performance duration
3MT	The 3-min all-out CP-test substitute

This review is dedicated to the memory of Dr. Henry Vandewalle who prematurely passed away in 2019. He was a highly knowledgeable and remarkably insightful scientist whose perspectives and well-considered opinions about critical power and related topics helped shape this review. Those and his unbiased critique are sorely missed.

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Introduction

In their landmark paper, mundanely-titled: “The work capacity of a synergetic muscular group”, Monod and Scherrer (1965) introduced the very elegant concept which only later came to be known as “critical power” (CP; Moritani et al. 1981), “critical torque” (CT; Burnley 2009), or “critical

speed/velocity” (CS; Wakayoshi et al. 1992). The fundamental concept was not new, having been raised earlier in relation to speeds and fatigue of racing animals (Kennelly 1906) and human athletic records (Hill 1925; Henry 1955). Laying the foundations for their 1965 publication were also the Monod and Scherrer’s own earlier works (Monod and Scherrer 1957; Scherrer and Monod 1960). The concept’s elegance and attractiveness was in demonstrating that there is a smooth ‘transition of power’ among the phospholytic, glycolytic, and oxidative energy sources and that the form of the declining power–time curve is, or approximates, a hyperbolic function. Consequently, it was reasonable to assume that the power output asymptote along the time axis represents the highest work rate that could be sustained (i.e., ‘critical power’) “for a very long time without fatigue” (Monod and Scherrer 1965), or the ‘heavy–severe exercise boundary’ as it is now often referred to Jones et al. (2010), Hunter et al. (2021), Kellawan and Tschakovsky (2014).

There was certainly a good theoretical justification for CP’s assumptions to take root. When the CP concept was introduced, initially only for discrete muscles or muscle groups (Monod and Scherrer 1965), and then for systemic, whole-body exertions (Moritani et al. 1981), the ‘anaerobic-threshold’ concept did not yet exist or was in its infancy. The maximal lactate steady state (MLSS) test, considered by many as the gold standard of ‘anaerobic-threshold’ determination, was introduced in a preliminary form only in 1985 (Heck et al. 1985), although MLSS’s observational foundation (not recognized by Heck et al.) was laid out already 49 years earlier by Bang (1936). Cast against this background, it can be easily appreciated why the CP concept was so alluring as an endurance index for the better part of its first 2 decades. Indeed, most eventual studies, comparing CP with actual endurance performance, found the two outcomes moderately-to-well correlated in a variety of activities (Wakayoshi et al. 1993; Kolbe et al. 1995; Vautier et al. 1995; Florence and Weir 1997; Smith et al. 1999; Kennedy and Bell 2000).

For the past 3 decades, however, numerous studies have shown the assumptions underlying CP determination to be inaccurate, as derived CP intensities could typically not be maintained for sufficiently long durations of ~30 min or longer (e.g., de Lucas et al. 2013; Dekerle et al. 2003; Sawyer et al. 2014). Other studies have shown CP to markedly overestimate MLSS (Pringle et al. 2002; Dekerle et al. 2003; Caritá et al. 2009; Greco et al. 2012; Mattioni Maturana et al. 2016), and CP’s associated blood $[La^-]$ has repeatedly been shown to exceed maximal sustainable values (e.g., Jenkins and Quigley 1992; Keir et al. 2015; Nixon et al. 2021). Myriad details and perspectives have been investigated and discussed and numerous confounding factors raised to account for the variability of measured CP and its MLSS overestimation (e.g., Hill 1993; Muniz-Pumares et al.

2017; Vandewalle et al. 1997). The overwhelming consensus premise has been that experimental ‘noise’ and procedural inconsistencies (e.g., lack of a standardized protocol) are the fundamental sources of the observed discrepancies.

This widely-held premise has likely been the main reason for why little or no effort has been directed at elucidating the root cause of the CP–MLSS discrepancy and, consequently, the fact that after over half a century since the introduction of the CP concept, no credible explanation exists. Avoiding this enigma altogether, it has in recent years been suggested that CP constitutes a distinct entity, fundamentally different from MLSS (Jones et al. 2019a). However, that revised form of CP is no longer compatible with the original classic concept as defined by Monod and Scherrer (1965) and thus leaves the enigma unresolved.

In adhering to the original underlying premise, namely that CP denotes the highest exercise intensity sustainable without fatigue, this review aims to suggest a new perspective and offer reasoned explanations for the evident discrepancy between theoretical and observed CP.

The CP–MLSS discrepancy

CP’s variability and reliability

Variability

The ability to compare CP values across different studies, or to rely on a given determination, is adversely affected by many factors, and is further hampered by the lack of a standardized protocol. Below are some of the main variability-contributing factors (see also reviews by Hill 1993; Muniz-Pumares et al. 2017; Vandewalle et al. 1997).

CP determination models Three most commonly used CP determination models are 2-factorial—the linear work–time ($W-t$) and power–1/time ($P-1/t$) relationships, and the hyperbolic power–time ($P-t$) relationship. Although the same set of data is used with each of the three models, CP values often differ considerably between any two models by as much as > 10% (Bergstrom et al. 2014; Bergstrom et al. 2021; Clark and Macdermid 2021; Dupont et al. 2002; Gaesser et al. 1990; Mattioni Maturana et al. 2018; Sawyer et al. 2014). Kordi et al. (2021) ‘solved’ the 3-model conflict by a compromise, selecting the model with the least overall error for each of their 22 cyclists.

If actual data indeed conformed to true linearity (in the $W-t$ and $P-1/t$ relationships), then (within experimental error) all 3 models would have resulted in identical CP determinations. DW Hill, in his 1993 review, claimed that “the existence of differences between estimates derived from the 3 different models can be taken as evidence that

there is a systematic error in the predicting data”. Attempting to address this issue, Morton, for example, suggested an empirically-based 3-parameter model (Morton 1996), as well as a bioenergetically-based one (Morton 2006). On the other hand, Vautier et al. (1995) and Vandewalle et al. (1997) proposed a 35-min-long determination trial and DW Hill (1993) suggested an even longer, 40-min trial, to partially alleviate some of the model's conflict. Attempts to use or modify these and other models to make practical sense of observational data are still ongoing (e.g., Leo et al. 2022).

Trial-duration effects Short trial durations render different CP values than longer ones (Bishop et al. 1998; Clingeffer et al. 1994; Jenkins et al. 1998; Mattioni Maturana et al. 2018; Triska et al. 2018; Vandewalle et al. 1997; Vautier et al. 1995). Clingeffer et al. (1994) tested 8 kayakers in 1.5-, 4-, 10-, and 20-min trials and calculated CP by all possible combinations of 2 trials. CP, based on the two shortest trials was 43.9% higher than that derived from the two longest ones (214.5 vs. 149.1 W). Bishop et al. (1998) tested 10 young women in 1-, 2-, 3-, 4-, and 5-min randomly ordered cycling trials. CP based on the 3 shortest trials (1–3 min) was 22.6% higher than that derived from the 3–5 min ones (201 vs. 164 W). A similar pattern was also reported by Triska et al. (2018) despite very small differences in trial durations (2, 5, 10 vs. 3, 7, 12 min). Vautier et al. (1995) tested 10 young men for time to exhaustion (TTE) at 60, 73, 86, 100, and 120% of maximal aerobic power. CP was then calculated based on the two highest loads (100 and 120%) and then on all the loads whose TTEs ranged from 3.5 to 35 min. Mean exhaustion times at those two CP intensities were 29.0 and 48.6 min, respectively. While the former was shorter (more intense) than would be expected for MLSS, the longer-based CP very closely corresponded to the determined lactate steady state (68.8 vs. 68.7% $\text{VO}_{2\text{max}}$).

Although trial durations are central to CP-determination, neither protocols nor durations have ever been standardized, as is evident by the wide variety of employed durations and protocols reported in the literature. Consequently, duration variability has resulted in increased CP variability, thus adversely affecting inter-study comparability.

Endurance time variability Endurance times, whether CP-related or not, are notoriously variable and prone to fluctuating motivation, habituation, and training/learning effects in repeat testing or subsequent trials (e.g., Hill 1993; Morton 1994). TTE variability is usually more significant in longer exercise durations. For example, Poole et al. (1988) found 2–4% variation in ~4-min TTEs, but 4–6% in ~8-min efforts. McLellan et al. (1995), testing 15 men for TTE at 80% $\text{VO}_{2\text{max}}$, in 5 separate trials, found group TTE to increase from 14.4 to 18.2 min between trials 1 and 5. Individual variation, as high as > 31%, prevented this large difference

from reaching statistical significance. Morton (1994), in promoting his short ramp-test protocol for CP determination, objected to long test bouts, noting that “factors such as motivation or other psychological variables may have a role in determining the onset of exhaustion. That is, the end point is greatly influenced by voluntary effort and is therefore not so well defined”. Faude et al. (2017) found much greater variability in considerably longer TTEs. They determined MLSS in 25 men and then tested their TTE twice at that intensity. Although the mean TTEs of 50.8 and 48.2 min were rather similar (~ 5% difference), ICC was a low 0.62 with a remarkably high within-subject variance of 40.6%.

Hinckson and Hopkins (2005) had 8 competitive runners perform a series of 3 runs at fixed running velocities producing TTEs of approximately 2-, 4-, and 8-min. Those were repeated 5 days, 7 weeks and 14 weeks later. Repeated-measures analysis showed within-runner variabilities to increase with duration: 9.2, 13.0, and 16.0%, respectively. Although the authors considered TTEs to be reliable, they noted in their conclusion that conversion of TTE changes, at single fixed speeds, to changes in time-trial times “are model and individual specific and, therefore, only approximate”.

More equivocal findings of test-retest reliability were recently reported by Succi et al. (2021). While CP's ICC was a respectable 0.90, ICCs of the 4 test trials (85, 90, 95, 100% $\text{VO}_{2\text{peak}}$) ranged from 0.04 to 0.81, coefficients of variation ranged between 7.9 and 23.0%, and the coefficient for CP was 6.1%. On the other hand, MacInnis et al. (2019) found 20-min efforts of well- to highly trained cyclists to be somewhat less variable than 4 min efforts ($r^2 = 0.97$ vs. 0.95, respectively). A likely reason for this is that road cyclists are typically more accustomed to long, measured efforts, such as time-trials, that are typically in the 20- to 60-min range.

As TTE variability, at any duration, directly impinges on the reliability of CP determination, many researchers prefer maximal self-paced power outputs in fixed, predetermined durations (Clingeffer et al. 1994; Karsten et al. 2015; Kordi et al. 2021; Parker Simpson and Kordi 2017; Valenzuela et al. 2021).

Cadence variability Freely chosen cycling cadences are often much higher than those energetically most economical. Marsh and Martin (1993), for example, found self-selected cadences at 200 W to be ~ 30 rpm higher than the most economical cadence, in highly trained and comparably fit cyclists and non-cyclists alike (~ 90 vs. ~ 60 rpm, respectively). Exceptionally, in well-trained competitive cyclists riding their race bikes on a treadmill at ~ 82% $\text{VO}_{2\text{max}}$ intensity, Hagberg et al. (1981) found the preferred and most economical cadences to coincide at ~ 91 rpm. Hansen et al. (2021) claimed self-chosen cadence to depend on the preceding cadence, such as a previous trial. To complicate matters further, both optimal and preferred cadences increase

with rising power outputs (Ansley and Cangle 2009; Coast and Welch 1985; MacIntosh et al. 2000). The different cadences, whether self-selected or imposed, present different cycling efficiencies (Ansley and Cangle 2009; Coast and Welch 1985; Driss and Vandewalle 2013; Francescato et al. 1995; Foss and Hallén 2004; MacIntosh et al. 2000), as well as different maximal power outputs (e.g., Beneke and Leithäuser 2017). Consequently, non-optimal cadences will always result in lower power outputs, relative to metabolic cost, than could otherwise be produced (see also the distinction between most efficient and most powerful cadences, discussed later under ‘Which Optimum?’). This, in turn, will affect the calculated CP. Barker et al. (2006), for example, determined CP in 11 sprint and endurance runners twice, at 60 and 100 rpm, and found the faster, less efficient cadence to elicit 8.7% lower CP than the slower, more economical one (189 vs. 207 W, respectively).

Since optimal cadences, in terms of both economy and power output, change with different power outputs, actual CP testing is bound to be affected by these differential cadence effects.

Trial spacing Trial-duration effects notwithstanding, close-spaced trials (e.g., 4, 6, 9 min) will have greater variability of calculated CPs than more widely spaced trials (e.g., 3, 8, 15 min) due to the degree of uncertainty surrounding each measurement point (power output and TTE).

Pacing strategies A 3-min all-out substitute for conventional CP determination (3MT), introduced by Burnley et al. (2006), has occasionally been used in pursuit of a simplified, single-session CP protocol (e.g., Black et al. 2014; Clark et al. 2018; Constantini et al. 2014). It has been criticized, e.g., for lack of sensitivity (Bartram et al. 2017), but has not been endorsed as a pacing strategy for conventional CP protocols. The reason for this has not been discussed but likely lies in true all-out performance being impossible to maintain or ascertain in extended efforts beyond pure sprinting. It is questionable whether even efforts as short as the 30-s Wingate Anaerobic Test (Ayalon et al. 1974; Bar-Or 1987) can be performed in true all-out fashion. Consequently, nominal ‘all-out’ efforts are (aside from the omnipresent issue of motivation, see below) subject to variability due to individual pacing choice and fluctuations. With the advent of electromagnetically braked cycle ergometers, it has become much easier to employ constant work rates (CWR) (e.g., McLellan et al. 1995; Muniz-Pumares et al. 2017; Pallarés et al. 2020; Vanhatalo et al. 2011), or self-paced, time-trial-type efforts (Denadai et al. 2005; Garatachea et al. 2006; Kordi et al. 2021; Lillo-Beviá et al. 2022). Importantly, self-paced trials typically produce greater mean power outputs (or velocities) and thus higher CP values. For example, Black et al. (2015) found mean self-paced CP of 265 W to

be 6.7% higher than the corresponding CWR CP (250 W). Laursen et al. (2007), on the other hand, found TTE superior to time trial in 1500m but inferior in 5km runs. TTE's test-retest reliability was, however, always inferior.

Thus, employment of different pacing strategies can have a profound effect on CP and its variability between studies.

Motivation While absolute motivation is inherently assumed in CP testing, it is hard to expect 100% motivation in any one effort and particularly in the several trials required for CP determination. Indeed, in his review, DW Hill (1993) recognized that “large numbers of trials may be associated not only with changes in learning, but changes in motivation”. Thus, relative to the theoretical concept, less than absolute motivation adversely affects both power outputs and TTEs throughout the range of trial durations (see also under ‘Endurance Time Variability’, above).

Test–Retest reliability

When the factors of variability remain constant, as in test–retest conditions, CPs are typically rather repeatable with good to high test-retest reliability. In 8 trained cyclists, Macinnis et al. (2019) demonstrated test-retest CP reliability of $r^2 = 0.93$ and ICC = 0.98. Karsten et al. (2015) test-retested 11 moderately trained competitive cyclists in 3 time-trial versions. Despite the field conditions, the ICC values remarkably ranged from 0.96 to 0.99 (0.98 mean). In laboratory testing, however, Succi et al. (2021) found a somewhat lower test-retest ICC of 0.90.

CP overestimates MLSS

Direct CP–MLSS comparisons

The vast majority of studies comparing CP to MLSS or its proxies, or to direct endurance performance, found CP to markedly overestimate MLSS, or what is now often termed maximal metabolic steady-state (MMSS, e.g., Caen et al. 2021; Iannetta et al. 2021).

McLellan and Cheung (1992) compared IAT [Individual Anaerobic Threshold (Stegmann et al. 1981), a typically close proxy of MLSS/MMSS] in 14 trained and untrained cyclists. IAT was found to be 30 W/12.8% overestimated by CP. Although not questioning CP's conceptual validity, the authors concluded that “the methods used to determine CP overestimate the metabolic rate associated with a maximal steady-state La and acid-base response”.

In a recent very informative systematic review, Galán-Rioja et al. (2020) compared CP to MLSS and three ventilatory thresholds in 9 studies, 5 of which reported 6 CP–MLSS comparisons—3 of untrained volunteers (Greco et al. 2012; Pringle et al. 2002; Mattioni Maturana et al. 2016), and 3

of trained cyclists (Dekerle et al. 2003, Caritá et al. 2009; Greco et al. 2012). There was no significant difference between the trained and untrained cyclists, but as can be seen in Fig. 1, CP invariably overestimated MLSS, on average by 30.1 W/11.7% (range 20–39 W, 8.6–23.2%; $p = 0.0002$).

A study such as that of Keir et al. (2015), who reported CP to be only 3 W/1.3% higher than MLSS, was excluded from Galán-Rioja et al.'s analysis for failing to report CP's correlation with MLSS. A missing correlation raises the suspicion that, although CP and MLSS means were exceptionally close, they may have been only weakly correlated and possibly affected by different factors. An example of that is Smith and Jones (2001), who reported running CS that was only 4.3% higher than MLSS velocity, but with a 0.62 CS–MLSS correlation. Due to very large variance in the two variables, neither the difference, nor the correlation, proved statistically significant, forcing the authors to conclude that “the extent of disagreement is too great to allow one variable to be estimated from another in individual subjects”. Denadai et al. (2005) also found mean CS of 12 soccer players to be higher than MLSS (9.9%; 14.42 vs. 13.12 km h⁻¹) with $r = 0.90$ between the two. Close inspection, however, revealed that the high correlation was solely due to a single outlier who was ~2–4 km h⁻¹ faster than all other 11 players. The authors concluded that “CV [CS] does not represent a sustainable steady-state exercise intensity”.

In concluding his CP-concept review, DW Hill (1993) expressly stated that “CP is above the anaerobic threshold” (approximately the threshold identified by MLSS). Mattioni Maturana et al. (2016), despite finding the smallest

CP–MLSS discrepancy in Galán-Rioja et al.'s review (8.6%), concluded that the discrepancy “questions the ability of CP measures to determine the maximal physiological steady-state”. They further stated that there is a need for a “consistent testing method for accurate determination of CP”.

Not surprisingly, the persistence of such discrepant findings has already earlier prompted the question: “What is being measured and why?” (Dotan 2014).

Time to exhaustion

The vast majority of studies investigating CP TTEs, found CP intensity to be too hard to sustain for MLSS-comparable durations. Thus, while MLSS exercise intensity can, by definition, be maintained for over 30 min and even up to ~1 h, observed endurance times for CP intensities have typically been found in the 15–25-min range (e.g., de Lucas et al. 2013; Dekerle et al. 2003; Sawyer et al. 2014). Notable exceptions of just under 30 min, are Vautier et al. 1995 and Brickley et al. 2002.

By hyperbolic-function definition, the P – t curve meets its asymptote only at infinity. In practical terms, however, when the appropriate power and duration data are provided, the time for the hyperbolic power curve to converge on CP within a given margin can be calculated. For example, data by de Lucas et al. (2013) rendered a time of 1.50 h to come to within 1% of the 295 W CP (297.6W). Actual TTE at CP was, however, only 22.9 min although 309 W ought to have been sustainable at that duration. Times to 101% CP for Kordi et al. (2021) and Housh et al. (1989) were 2.00 and

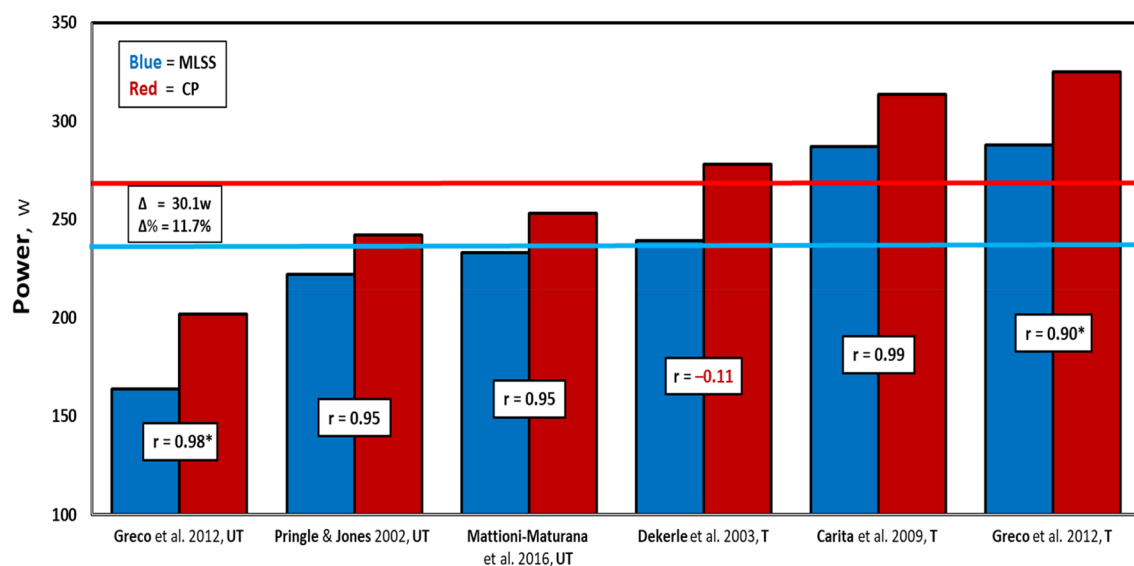


Fig. 1 Six CP–MLSS comparisons in trained (T) and untrained (UT) cyclists as reported in 5 studies and summarized in a review by Galán-Rioja et al. (2020). Although mean CP (red) is significantly higher than mean MLSS (blue) (30.1 W/11.7%, $p = 0.0002$), the

typical correlation between the two is very high ($r = 0.90$ – 0.99), suggesting that they both reflect a common factor (e.g., endurance). * - different cohorts in the same study

2.19 h, and the predicted sustainable 30-min power outputs were 9 and 13 W (3.7 and 4.6%) above CP, respectively. Thus, it is not only that, in actual testing, CP intensity cannot be maintained for the ‘predicted’ ~ 2 h, it cannot be maintained even for the MLSS-compatible 30 min where a yet-higher power output ought to have been sustainable.

The findings of Hughson et al. (1984) in competitive runners provide another perspective of the same issue. The authors reported CS-predicted 10-km running times to be 2–3 min (~ 5–9%) shorter (faster) than the actual times the runners could attain in a 10-km race (32:16–36:16 min). Moreover, the correlation between the predicted and actual running velocities was only moderate ($r = 0.67$), explaining < 45% of the observed variance.

CP–MLSS correlation Despite the magnitude of the CP–MLSS differences, there typically are moderate to high correlations between the two, attesting to CP being closely related to endurance capacity. It is unclear why Dekerle et al. (2003) found no correlation ($r = -0.11$), but in the other 5 comparisons in Galán-Rioja et al.’s (2020) review (Fig. 1), the mean CP–MLSS correlation quotient was 0.954, strongly suggesting that similar factors determine both.

The heavy–severe exercise boundary

The term ‘anaerobic threshold’ has long been used to denote the “intensity of exercise, involving a large muscle mass, above which measurement of oxygen uptake cannot account for all of the required energy” and “the exercise intensity above which there is a net contribution of energy associated with lactate accumulation” (Svedahl and MacIntosh 2003). In other words, it is the highest exercise intensity, or metabolic rate, at which a prolonged equilibrium can be achieved between lactate production and its elimination. The term ‘Anaerobic’, however, has come under scrutiny in recent years (Ferguson et al. 2018; Brooks 2021) because, historically, it had been coined to describe environments deprived of oxygen, rather than metabolic processes that require no oxygen to be carried out (i.e., glycolysis and phospholysis). In current terminology, therefore, the ‘anaerobic threshold’ of old is now often referred to as “maximal metabolic steady state” (MMSS). Conceptually, MMSS is no different than MLSS, but the latter is typically taken to denote the result of the imperfect MLSS test rather than the precise maximal steady state it attempts to determine.

MLSS’s strengths and shortcomings

There are very few perfect tests, if any, and the MLSS test is certainly not one of them. Its two main shortcomings in terms of accuracy are, first, being a bracketed test, whose resolution is limited by the magnitude of the stage-to-stage

incrementation. Secondly, it is a multi-session test, which aside from making the MLSS test highly impractical, also renders it susceptible to day-to-day fitness fluctuations (e.g., Dotan 2012). While these shortcomings may dent MLSS’s practical accuracy, they do not detract from its conceptual validity as a systemic, whole-body measure of the upper limit of sustainable metabolic rate (MMSS). The fact that even minor increases of power output beyond MLSS are promptly followed by increased $[La^-]$ and diminishing TTEs is a robust justification of the concept.

Admittedly, MLSS’s very cumbersome nature has greatly limited its practical use (e.g., in sports testing) as well as in research-oriented applications. Furthermore, when MLSS is being used, it is often limited to 3 stages (Keir et al. 2015; Kordi et al. 2021; Valenzuela et al. 2021; Wakayoshi et al. 1993; Hinckson and Hopkins 2005), thus further compromising its accuracy. Nevertheless, studies have shown $MLSS_{power}$ (unlike $MLSS_{La}$) to be highly reliable, e.g., test-re-test ICC = 0.985 (Adam et al. 2015) and ICC = 0.98 with 3% coefficient of variance (Hauser et al. 2013). However, MLSS’s gold-standard status is not only based on its reliability and sound physiological concept, but also on validations against actual endurance performances. For example, Harnish et al. (2001) ($r = 0.85$ and 0.92 with 5- and 40-km cycling time-trials, respectively), Haverty et al. 1988 ($r = 0.87$ with 5-km running time-trial), and Jones and Doust (1998) ($r = 0.92$ with 8-km simulated running race).

Novel claims

Until recently, studies typically reported CP values under various conditions or activity modes, quantified CP’s differences from MLSS and other threshold criteria, or tried to formulate practical rules by which to make CP more MLSS-compatible. This changed in 2019 with the convention-challenging review: “The maximal metabolic steady state: redefining the ‘gold standard’” (Jones et al. 2019a). The review centred around the claim that CP, not MLSS, is the gold standard for MMSS determination, insisting that “comparisons between MLSS and CP, undertaken in the mistaken belief that they are theoretically synonymous... [should be] discontinued”. This was surprising in that it contradicted the authors own earlier assertions that “MLSS, and not the CP, represents the upper limit of the heavy exercise domain in cycling”, or that “the boundary between the heavy and severe domains is given by the maximal [lactate] steady state” (Smith and Jones 2001; Pringle et al. 2002; Burnley et al. 2006).

These claims found support by Galán-Rioja et al. (2020), but a scientific debate ensued (García-Tabar and Gorostiaga 2019, 2021; Jones et al. 2019b; Galán-Rioja et al. 2021, Keir et al. 2021) with no ‘declared winner’. However, being revolutionary and coming from the highly reputable and most

prolific group in critical power research, the authors' arguments and views carry much weight. It is thus imperative to critically scrutinize the arguments as they pertain to this review's theme.

As has already been shown, typical TTEs at CP intensities are in the ~ 15–25-min range. This means that observed CP is incongruent not only with MLSS but, more importantly, with the fundamental Monod and Scherrer's concept, defining CP as indicating the maximum work rate that can be kept up "for a very long time without fatigue" (Monod and Scherrer 1965).

Therefore, if it is argued that observed CP is a generally valid measure of the heavy–severe exercise boundary and distinct from MLSS (i.e., lying above the highest lactate equilibrium), then there is no need to denigrate MLSS and rather just focus on CP's virtues. If, on the other hand, it is claimed that MLSS is conceptually valid but a gross CP under-estimator in practice, then these claims ought to be closely examined.

MLSS's methodological concerns As has already been noted, MLSS, like any other physiological tests, is not perfect. Yet, the MLSS shortcomings noted in the novel review, come short of accounting for the typically observed CP–MLSS difference. For brevity, only the most significant of the cited shortcomings will be addressed here.

As noted earlier, MLSS's largest single source of potential MMSS underestimation is its bracketed nature, whereby MLSS is defined as the last stage before the one in which $[La^-]$ rise exceeds 1 mM in the test's last 20 min. The review misinterprets this to mean that a 1-mM $[La^-]$ rise is what defines MLSS attainment and if $[La^-]$ rose less than that in the penultimate stage then it constitutes MLSS underestimation. This is not the case. The 1-mM criterion (Beneke and von Duvillard 1996) was introduced only to allow for determination uncertainty and defines an upper limit of accepted deviation from true steady state, which theoretically should have been zero $[La^-]$ rise. Thus, a stage that has shown, for example, a 0.1-mM rise is, in principle, already an MLSS overestimation—clearly not an underestimation thereof.

Even when adhering to the authors' definitions, the highest possible mean MLSS underestimation would be no more than half the test's resolution (stage-to-stage incrementation). Using Galán-Rioja et al.'s (2020) six comparisons (Fig. 1) as a representative sample, mean incrementation was 11.5 W. Maximal mean underestimation, therefore, could not have exceeded 5.75 W and since the mean CP–MLSS difference was 30.1 W, such underestimation could, at most, account for only < 20% of that. Other cited MLSS shortcomings are even less significant and will not be discussed here. Thus, MLSS inaccuracy is not nearly as significant as that of CP and cannot be used to explain the consistently-observed CP–MLSS differences.

Delayed steady-state and muscle–blood $[La^-]$ equilibrium In support of observed-CP's validity, CP's apparent lack of equilibrium is ascribed to "delayed steady-state vs. non-steady state behavior". It is correctly noted that the time to metabolic plateau (e.g., of VO_2 , or $[La^-]$) increases with increasing exercise intensity. This, however, does not explain how the intensity associated with a presumed index of endurance cannot be sustained long enough to attain such equilibrium. Physiologically, the assertion's main weakness lies with its time scales. Dotan et al. (2003) measured $[La^-]$ in men and boys, during recovery from the Wingate Anaerobic Test. Impedance of La^- release from the muscles and the resultant delay in attaining peak blood $[La^-]$ were expected to be near maximal under the conditions of high muscle $[La^-]$ and $[H^+]$ following the Wingate Anaerobic Test. Although the 30-s all-out test is > 3-fold as intense as MLSS or CP, the mean time lag to post-test peak $[La^-]$ was 7.6 min for the men and only 5 min in the boys. $[La^-]$ equilibrium in sub-maximal constant-power efforts, such as in MLSS testing, is attained in ~ 10 min or less.

Even more directly to the point, Jorfeldt et al. (1978), whom the authors cite, showed no $[La^-]$ difference between the 4th and 12th min of constant-power efforts at 29, 51, and 71% VO_{2max} (the latter being the approximate MLSS/CP intensity). A significant difference finally materialized only at 87% VO_{2max} . As this intensity is above that of MLSS or CP, it is of no consequence to the issue of unattained or delayed equilibrium. Thus, if equilibrium cannot be attained in 20- or 25-min of exercise at CP intensity, then this is not due to delayed response but rather to work-rate unsustainability.

Muscle- vs. Blood- $[La^-]$ equilibrium Both Jones et al. (2019a) and Galán-Rioja et al. (2020) claimed CP to reflect muscle- rather than blood–lactate equilibrium, based on findings such as those by Jorfeldt et al. (1978) which, as shown above, are inconsequential to intensities relevant to the CP–MLSS controversy. The notion of muscle $[La^-]$ equilibrium is hard to follow even as a conceptual construct. Unlike critical torque or force (e.g., Monod and Scherrer 1965), whole-body exercise involves multiple muscles working together but at vastly different intensities. Even at MLSS/CP intensity, highest $[La^-]$ equilibrium, or any other equilibrium for that matter, may be true for a given muscle, but other muscles are, concurrently, net La^- producers, La^- consumers, or inactive. Thus, if CP is to represent systemic, whole-body endurance capacity, one would need to monitor all muscles to obtain a composite CP value, which is both impractical and presently outright impossible. Within experimental error and a 20- to 30-min time frame, the composite muscle $[La^-]$ value would likely be indistinguishable from blood $[La^-]$. Furthermore, to associate a composite muscle $[La^-]$ with a composite power output, each mus-

cle's performance would have to be monitored individually. Thus, the notion of muscle- $[\text{La}^-]$ equilibrium in a systemic context is an unsupportable premise.

In a more recent study by the same group (Nixon et al. 2021), 10 well-trained runners ran for 30 min $0.5 \text{ km}\cdot\text{h}^{-1}$ above MLSS velocity ($15.2+0.5 = 15.7 \text{ km}\cdot\text{h}^{-1}$), raising their final-20-min $[\text{La}^-]$ by 4.4 mM. Although completion of that task and being able to maintain VO_2 steady-state attests to the runners' high fitness level, their fast-rising $[\text{La}^-]$ proves that metabolic steady-state was not attained. Noteworthy is that even 100% $\text{VO}_{2\text{max}}$ can be maintained for several minutes, especially by well-trained athletes, but that clearly does not indicate a steady-state. Second, according to the authors' own underestimation claims, a $0.5 \text{ km}\cdot\text{h}^{-1}$ MLSS testing incrementation would mean an average of $\sim 0.25 \text{ km}\cdot\text{h}^{-1}$ MLSS underestimation, which then would mean MLSS of $\sim 15.45 \text{ km}\cdot\text{h}^{-1}$ (rather than 15.2). Thus, a mere $0.25\text{-km}\cdot\text{h}^{-1}$ (1.6%) velocity increment (from 15.45 to $15.7 \text{ km}\cdot\text{h}^{-1}$) caused a steep 4.4-mM $[\text{La}^-]$ rise. Contrary to the authors' intentions, these findings lend extra credence to the

imperfect MLSS test rather than to the novel CP concept the authors were trying to promote.

Conflict resolution?

CP's conceptualization, as defined by Monod and Scherrer in (1965), has not changed in most researchers' minds throughout the years (e.g., Hill 1993; Vandewalle et al. 1997; Keir et al. 2018). In his widely regarded review, DW Hill (1993) stated that at least “*theoretically*, the power asymptote of the relationship, CP..., can be sustained without fatigue”, while Keir et al. (2018) stressed that “The CP concept is predicated on the hyperbolic relationship that exists between exercise intensity and tolerable duration... [and that] from a *mathematical standpoint*, such a relationship implies that exercise performed constantly at the asymptotic intensity should be tolerated indefinitely. That is, it has been well and widely recognized that despite observational discrepancies, CP's conceptual validity has remained unchanged.

Thus, the newly-proposed CP concept, with CP intensities unsustainable for more than $\sim 15\text{--}25$ min, is clearly incompatible with the original concept predicated on the asymptotic indication of prolonged-endurance capacity. However, although the abyss between the two concepts is unbridgeable, the conflict may be resolved by recognizing the possibility that beyond the heavy–severe exercise boundary (MMSS), conceptually defined by MLSS and CP, there is yet another threshold or boundary, which presumably is what the novel concept's promoters refer to. Such an entity could, for example, be termed ‘ CP_2 ’, not unlike the distinct ventilatory differentiation between VT_1 and VT_2 , but it would require further research to be validated.

The following data and discussion will present what, beyond experimental noise or biological variability, the likely reasons are for the observed CP–MLSS difference. Had this been shown and accepted earlier, the need for an alternate CP concept would likely have not been perceived.

Why is observed CP unsustainably high?

The ‘Exclusion Zone’

In their landmark publication, Monod and Scherrer (1965) noted, regarding isolated muscles or muscle-groups, that “the relation between W_{lim} and t_{lim} is really linear only for tests whose maximum time [bout duration] exceeds 5 minutes in the case of static work, and two minutes in the case of intermittent static work”. That is, the observed force or torque in maximal efforts, shorter than the specified durations, was lower than expected, based on the linearity of the $W\text{--}t$ relationship. Similar observations were eventually also made in power-based studies, prompting many researchers

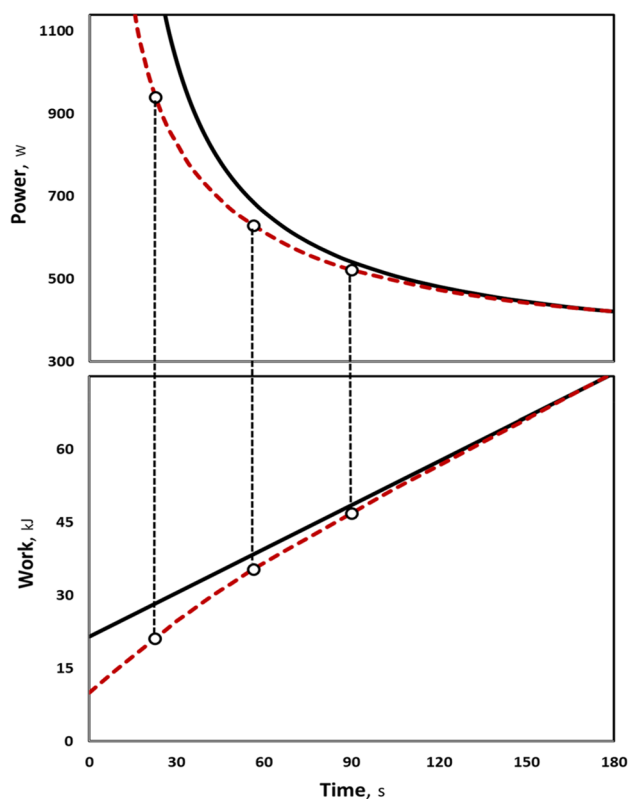


Fig. 2 Schematic representation of the ‘exclusion’ zone’s non-linear distortion (dashed red line) of the theoretically linear $W\text{--}t$ relationship (solid black line), in the lower plot, and its corresponding depressing effect (dashed red line) on the hyperbolic $P\text{--}t$ curve (solid black line), in the upper plot. The actual extent of such distortions can greatly exceed the 180 s of this example (adapted from Bergstrom et al. 2021)

to avoid or exclude trials shorter than 2 min and to have their first, most intensive trial at ~ 3 min or longer. For example, in their remarkable review, Vandewalle et al. (1997) referred to a 3.5-min zone, where observed cycling values under-represented the expected linearity of the W - t relationship and recommended exclusion of trials shorter than that. It is noteworthy that the 3.5-min (210s) time point is roughly the demarcation separating the ascending trajectory portion of the power-time (P - t) curve from that of the horizontally trajected portion (see Figs. 3, 4 and 5). Figure 2 is a schematic representation of how the non-linear distortion of the W - t relationship inside the ‘exclusion zone’ causes a depression of the hyperbolic P - t curve within that time frame.

Not unlike AV Hill’s long-celebrated force-velocity hyperbolic relationship (Hill 1938), much of the attractiveness of CP’s P - t relationship lies in its seamless transition from extreme phospholytic/glycolytic to fully oxidative energy sources in a mathematically-elegant hyperbolic manner. It is true, as is the case for AV Hill’s force-velocity relationship, that the very extremes of power and duration are physically or physiologically unattainable. Thus, unless one abandons the CP concept’s hyperbolic underpinnings, there is no apriori, conceptually compelling reason to claim that the fidelity of the hyperbolic P - t relationship could not

be maintained throughout the range of several seconds to well over an hour. It is thus conceptually troubling that such a significant portion of the hyperbolic P - t relationship is discarded. It is further disconcerting since exclusion of the first 2–5 min of the P - t relationship practically eliminates all exercise intensities in which non-oxidative metabolism is the predominant energy source. This undercuts the universality and conceptual attractiveness of the CP concept, is akin to discarding some 50% of AV Hill’s force-velocity relationship, and questions the very validity and usefulness of the CP concept. Thus, rather than an apriori exclusion, the reason for this apparent aberration ought to be elucidated.

Conceptual essence of the CP–MLSS discrepancy

Shorter stages—Higher CP

When the P - t curve is derived from data obtained from within the ‘exclusion zone’ (e.g., 1, 2, 3 min) vs. those obtained from longer-duration trials (e.g., 5, 9, 15 min), it is readily apparent that the shorter-duration power values fall significantly below what is expected based on the longer trials. This phenomenon has been discussed earlier and is well documented or can readily be derived (e.g., Bishop

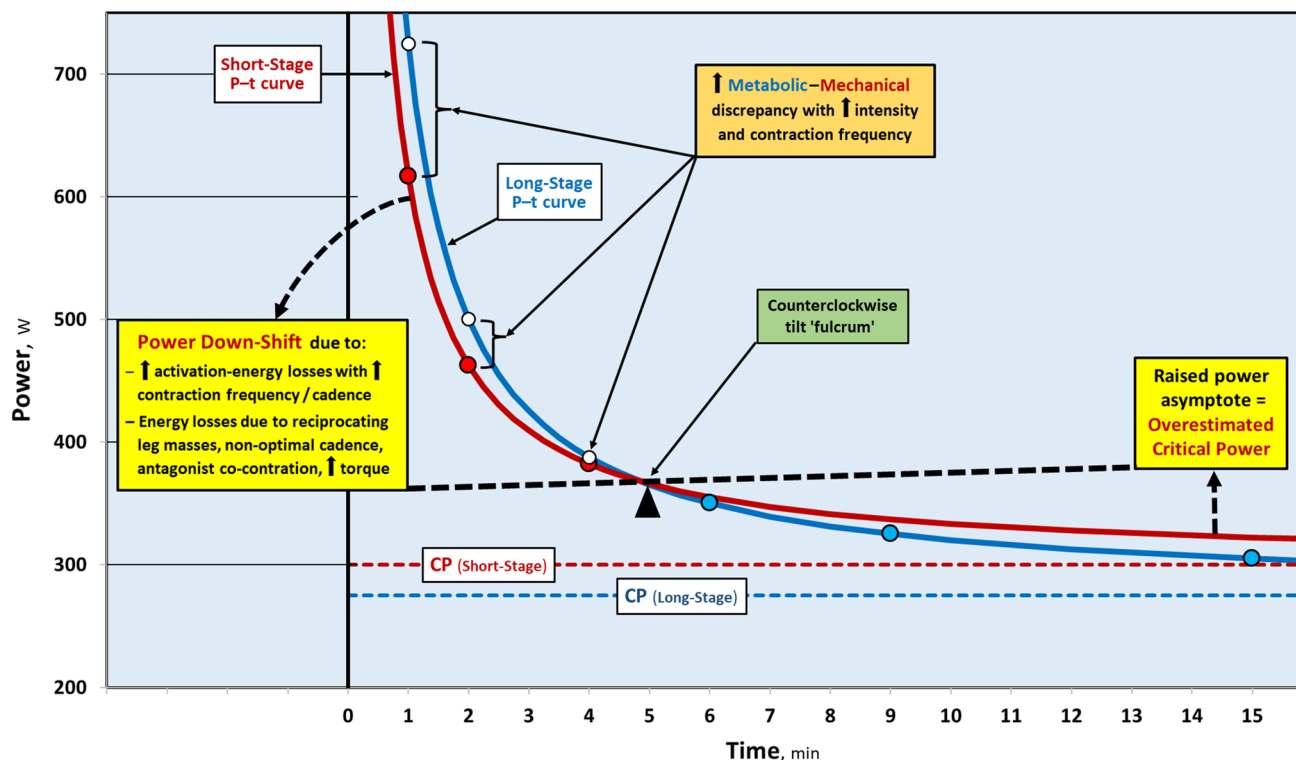


Fig. 3 Conceptual illustration of the effect under-par power output values of a short-trial CP test (red markers) have on the power asymptote and estimated CP, relative to a long-trial test (blue markers)—a 25-W CP over-estimation (300 vs. 275 W) in this schematic exam-

ple. White markers are the power output values of the long-trial P - t curve, corresponding to the durations of the short-trial values, and showing exponentially increasing disparity with decreasing trial durations and increasing contractile intensity

Fig. 4 The power–time curve (solid red line) based on 180-, 300-, and 720-s trials (red markers). The 1-s instantaneous power output (blue marker) is a small fraction of the predicted power output for a 1-s supra-maximal effort (data by Kordi et al. 2021). The empirical solution to this discrepancy (the 3-parameter model proposed by Morton 1996) was to shift the X-axis leftward. In this example, a 25-s shift brought the curve (blue line) to match the observed 1-s power output

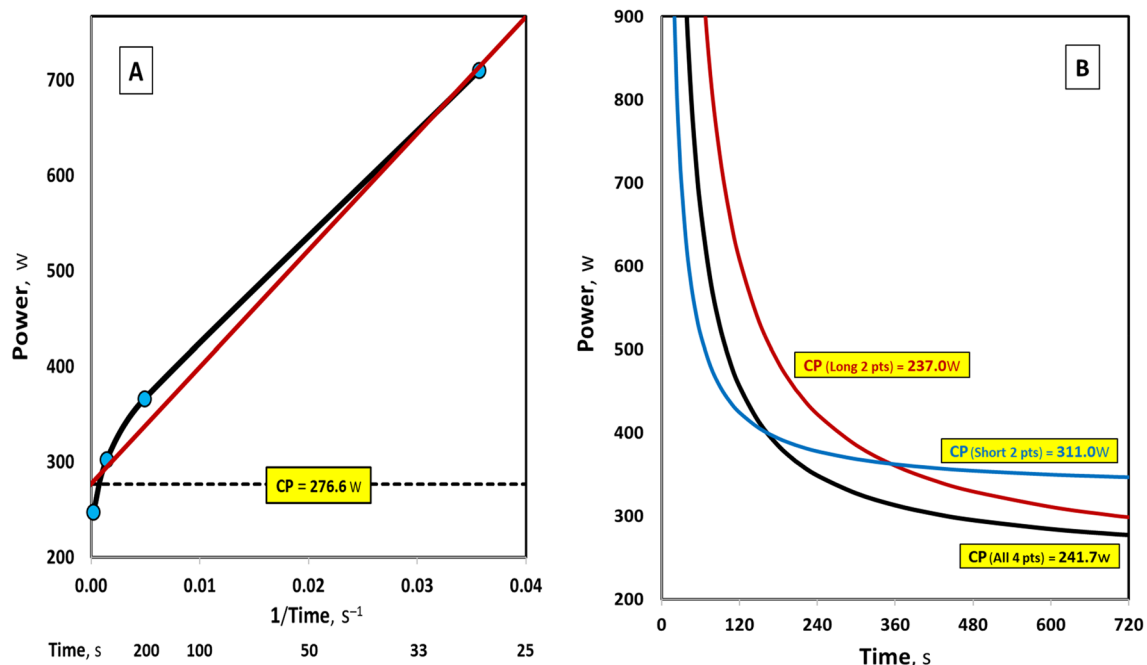
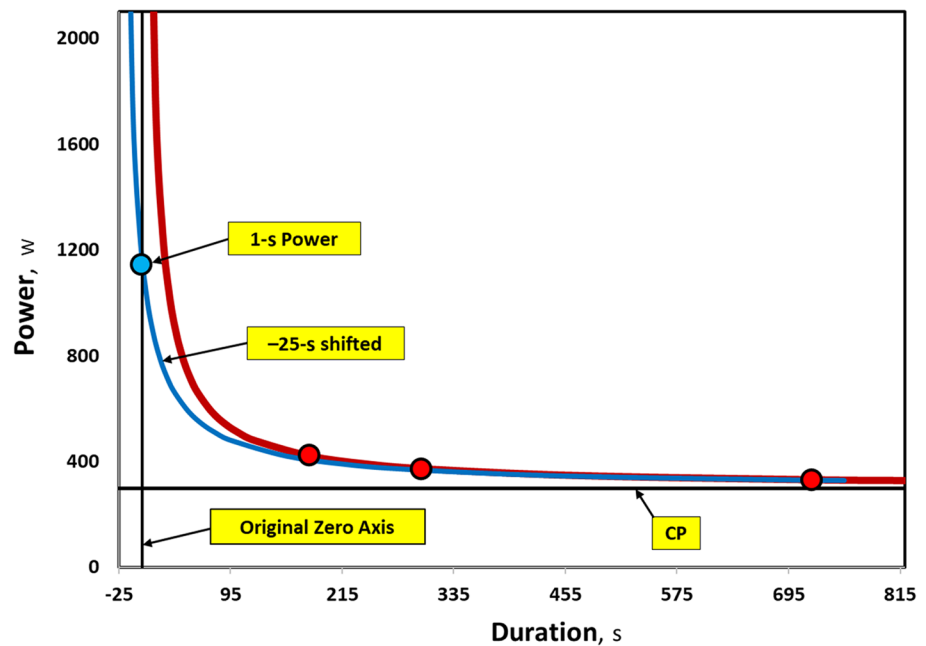


Fig. 5 Effects of forcing linear models upon non-linear data (Pallarés et al. 2020). **A** The non-linear $P-1/t$ relationship (solid black line) and the linear regression (red line) force upon it. **B** Power–time curves and corresponding calculated CP values for all 4 reported data

points (black line), for the 2 shortest-duration points (blue line), and for the 2 longest-duration points (red line). Note large differences in predicted power outputs and the derived CP values

et al. 1998; Clingeleffer et al. 1994; Hill 1993; Jenkins et al. 1998; Mattioni Maturana et al. 2018; Triska et al. 2018; Vandewalle et al. 1997; Vautier et al. 1995). The direct consequence of this is that CP derived from the shorter trials is markedly higher than that based on the longer ones, as is conceptually depicted in Fig. 3.

Morton (1996), recognizing that all three 2-parameter models ($W-t$, $P-1/t$, $P-t$) overestimated CP, noted that “it is a recurrent finding that the experimentally determined CP seems to obviously overestimate the power output that can be maintained continuously” (a reiteration of statement made two years earlier; Morton 1994). To correct that,

Morton proposed a corrective 3rd parameter, whereby “a time asymptote significantly less than zero is obtained”. The correction makes experimental data better conform to the true hyperbolic model, as depicted in Fig. 4, based on Kordi et al.’s (2021) data (3-, 5-, and 12-min trials). A 1-s peak power output (~1125 W), greatly overestimated by the standard P – t relationship, is made to fit it by the 3rd parameter shifting the relationship leftward. Applying his correction to McLellan and Cheung’s (1992) data, Morton demonstrated a reasonable 20.1 W decrease in mean CP (from 265.1 to 245.0 W), but some individual corrections were clearly unreasonable. It should be emphasized that while Morton’s procedure may roughly correct power discrepancies, it is an empirical tool (a ‘corrective lens’) that does not address the discrepancy’s root cause.

Thus, short trials depress the observed short-duration power outputs relative to those expected based on longer trials, or on true W – t and P – $1/t$ linearity. As can be seen in Fig. 3’s schematics, the disparity increases with decreasing durations and increasing intensities. Thus, if CP is expected to approximate MLSS/MMSS, then long trials provide better estimates since their lower derived CP values are sustainable for durations that are more typical of MLSS (Hill 1993; Vandewalle et al. 1997; Vautier et al. 1995). Interestingly, Luttikholt and Jones (2022) compared three P – t relationships, based off three peak-power determinations via a wide range of incremental test protocols and stage durations (2–360 min). The resulting hyperbolic curves, centered on their peak 6-min values, showed the same ‘fulcrum’ effect (counter-clockwise shift) as already noted by Vandewalle et al. (1997) and depicted here in Fig. 3.

Whatever the reason for it might be, the considerable disparity between expected and observed power output in shorter, higher intensity efforts, must be considered the main, if not the sole reason for the observed CP–MLSS discrepancy. The likely underlying reasons are discussed hereafter.

Pseudo linearity

By definition, a perfectly linear W – t relationship also means a perfect linearity of its P – $1/t$ counterpart, and vice versa. Beyond the 2–3.5-min ‘exclusion’ zone, commonly observed in CP testing, presumption of linearity has typically been the working premise in CP research and deviations from it have generally been regarded as due to experimental noise, justifying and prompting the use of more than two trials in CP estimation. Had the linearity premise been valid, then any two of the typical 3–5 data points could be used interchangeably to derive singular or closely similar CP values.

Beyond the already cited examples of the differential effect of short vs. long trials, and those first summarized by

DW Hill (1993), the reported data of two different sample studies vividly demonstrate that observed deviations from linearity are much more fundamental than mere experimental errors. With one trial deep in the ‘exclusion zone’, Pallarés et al. (2020) had 14 endurance-trained cyclists complete 4 maximal efforts of 28, 203, 684, and 4446 s (~0.5–74 min, 710–247 W, respectively) (Fig. 5). The intercept of the 4-point P – $1/t$ linear regression indicates CP of 276.6 W, but it is glaringly apparent (Fig. 5A) that these 4 points do not reside on a linear relationship and that linearity would not be attained even had the shortest trial been omitted. This is not benign. When the same 4 points are plotted as a P – t relationship (Black curve; Fig. 5B), the derived CP (241.7 W) is 34.9 W (12.6%) lower than the P – $1/t$ -derived CP (276.6 W). Moreover, CP derived from the two shortest (28 s, 203 s) trials (311 W; blue line) is 74 W (31.2%) higher than that based on the two longest (684 s, 4446 s) trials (237 W; red line). Thus, depending on the relationship being used and the points chosen, 4 quite different CP values are derived.

While the magnitude of the discrepancies in Pallarés et al.’s study can be ascribed in part to the inclusion of a very short trial (28 s), a study by Jenkins et al. (1998) is instructive in demonstrating that distortion of linearity exists even when shortest trials are well outside the ‘exclusion zone’. Eight recreational cyclists performed 5 maximal trials averaging 412.5, 387.5, 362.5, 337.5, and 312.5 W, and lasting 252, 408, 618, 780, and 1542 s (4:12–25:42 min), respectively (Fig. 6). Again, it is abundantly clear that the P – $1/t$ relationship is curvilinear (Fig. 6A) with a derived CP (5-point linear regression) of 303.2 W. However, when plotted separately (Fig. 6B), the 3 shortest, all 5, and the 3 longest trials, produce three markedly different CP values of 278, 303, and 332 W, respectively—a range of 54 W and a 16.2% discrepancy relative to the highest estimate.

In a torque rather than power context, Pethick et al. (2020) measured TTEs of intermittent isometric contractions (6 s on, 4 s off) at different MVC percentages. Importantly, the shortest TTE was 6:12 min (quite longer than Monod and Scherrer’s 2–5-min ‘exclusion zone’). Nevertheless, CT based on the reported 4-trial torques was 59.6 Nm, while that based on the two shortest trials was 19.2% higher (71.2 Nm), and the longest-trial-based CT was 34.1% lower (39.3 Nm).

Clingeffer et al. (1994) performed this analytic exercise with their own data (4 cycle-ergometer time-trials of 1.5, 4, 10, and 20 min). They did not find any linear relationships and their 4-point best-fit-calculated CP was 163.6 W. They then calculated CP for all 6 two-point combinations of their data. Not surprisingly, the lowest CP, 149.1 W, was derived from the two longest trials, while the highest came from the two shortest trials, 214.3 W—a 65.2 W difference, 30.4% of the highest CP. Even when eliminating the short, 1.5-min

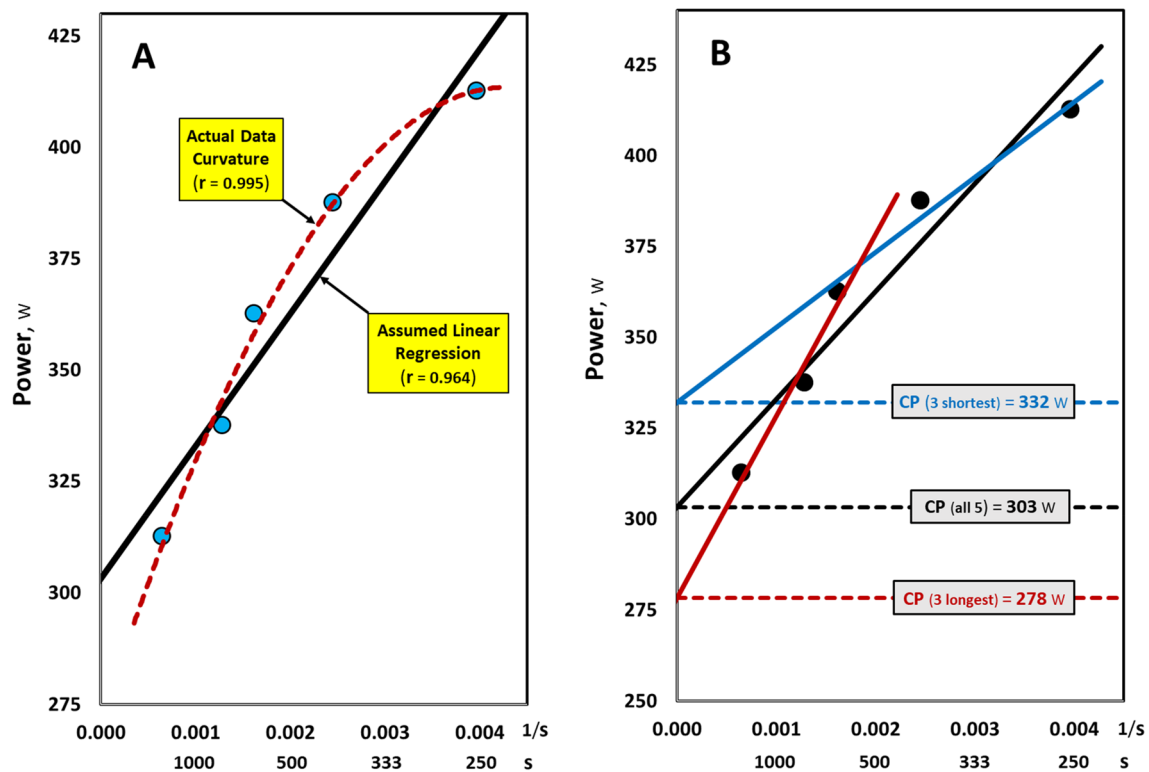


Fig. 6 A similar example to that of Fig. 5, based on data reported by Jenkins et al. (1998). Five data points were used, rather than 4, which did not improve linearity of the P – $1/t$ relationship (left) and rendered a CP value of 303 W (black regression lines at both left and right).

However, when using only the 3 longest or 3 shortest data points, the derived CP values were 278 and 332 W, respectively (a 54-W ambiguity zone)

trial, CP determined from the 4- and 10-min trial combination (172.9 W) was still 23.8 W (16%) higher than the 149.1 W derived from the 10- and 20-min combination.

A particularly illuminating example of the misleading nature of linearity assumptions is the study by Dupont et al. (2002). Nine young men were field-tested, running to exhaustion at 110, 120, 130, and 140% of their respective maximal aerobic velocities in an intermittent fashion (15 s on, 15 s off). Based on all 4 trials, determined CS was $4.82 \text{ m}\cdot\text{s}^{-1}$, but based on the 2 shortest and 2 longest trials, CS increased to 5.30 (+10.0%) or decreased to $4.66 \text{ m}\cdot\text{s}^{-1}$ (−3.5%), respectively. More extreme examples have already been presented, but peculiar to this example is that despite being a field test with two of its trials being only 1:40- and 2:47-min long, the reported correlation for the D–t (distance–time) relationship was one of seeming perfect linearity: $r^2 = 1$. Again, had the relationship been indeed linear, 3 identical CS determinations would have resulted.

When reported data permit, the non-linearity phenomenon can be demonstrated in nearly all other studies (Caritá et al. 2009; Clingeffer et al. 1994; Dekerle et al. 2003; de Lucas et al. 2013; Dupont et al. 2002; Greco et al. 2012; Hinckson and Hopkins 2005; Housh et al. 1989; Housh

et al. 1990; Jenkins et al. 1998; Karsten et al. 2015; Kirby et al. 2021; Kordi et al. 2021; Mattioni Maturana et al. 2016; Muniz-Pumares et al. 2019; Nixon et al. 2021; Pethick et al. 2020; Valenzuela et al. 2021; Vandewalle et al. 1997). Table 1 highlights the non-arbitrary nature of the phenomenon. The 30-, 60-, and 120-s high-intensity power outputs, determined based on the shortest trials, are compared with those derived from all or only the longest trials. Although the 20-study (21-cohort) sample contains two studies that do not conform to the general trend (Kordi et al. 2021; Vandewalle et al. 1997), the short-trial-based 2-min power outputs were, on average, 6.8% lower than those derived from all trials, and 18.4% lower than those based on the longest trials. Expectedly, the 1-min and 30-s discrepancies were greater yet, 14.6 and 21.3% relative to all trials, 27.6 and 34.7% relative to the long trials, respectively. All differences were highly significant ($p \leq 0.001$).

The mean duration of the shortest trials was 2:52.5 min. While short, ‘exclusion zone’ trials (<3.5-min), may have contributed to the degree of observed non-linearity, it ought to be noted that four of the nine largest discrepancies were associated with protocols that used four of the eight longest short-trials (Dekerle et al. 2003; Jenkins et al. 1998; Kirby

Table 1 Percentage of 30-, 60-, and 120-s short-trial-predicted power outputs, relative to those predicted from all- vs. the longest trials

Studies (All are cycling studies, unless otherwise indicated)	Short-est trial min:s	No. of trials ^a	Short-trial-based power outputs relative to those derived by all trials, or longest ones %					
			All trials ^a			Long trials ^a		
			30 s	60 s	120 s	30 s	60 s	120 s
Caritá et al. (2009)	3:58	3	99.6	98.0	96.4	78.8	83.1	88.1
Clingeffer et al. (1994) (kayaking)	1:30	4	67.1	77.6	90.8	43.1	53.5	69.0
Dekerle et al. (2003)	3:44.4	4	68.5	91.1	95.1	49.9	58.2	69.9
de Lucas et al. (2013)	3:48	3	94.7	89.1	98.2	86.2	90.1	94.3
Dupont et al. (2002) (running)	1:40	4	81.5	90.1	97.6	63.7	76.1	88.7
Greco et al. (2012) (trained)	3:47	3	90.5	93.7	96.9	74.3	81.0	88.6
Greco et al. (2012) (untrained)	4:27	3	95.6	96.8	98.3	88.2	91.2	94.6
Hinckson and Hopkins (2005) (running)	1:48	3	86.5	93.0	98.4	75.2	85.0	94.0
Housh et al. (1989)	1:41.6	5	53.8	61.8	73.8	33.0	39.7	50.8
Housh et al. (1990)	2:42	4	86.7	91.3	96.6	86.9	91.5	96.7
Jenkins et al. (1998)	4:12	4	69.6	76.4	85.2	58.3	65.9	76.5
Karsten et al. (2015)	2:23	3	55.3	71.3	89.0	42.9	58.6	78.4
Kirby et al. (2021)	5:00	3	34.7	56.1	91.4	16.8	28.7	51.5
Kordi et al. (2021)	3:00	3	106.3	104.2	101.9	111.0	107.4	103.6
Mattioni Maturana et al. (2016)	1:42	5	91.7	94.6	97.6	81.1	86.2	91.9
Muniz-Pumares et al. (2019)	2:15	5	91.7	94.7	97.7	82.1	87.4	93.0
Nixon et al. (2021) (running)	3:05	4	63.9	76.6	88.9	50.9	64.9	80.3
Pallarés et al. (2020)	0:28	4	63.9	77.1	92.9	40.9	52.9	69.8
Pethick et al. (2020) (torque)	6:12	4	63.1	67.0	73.7	36.4	39.9	46.4
Valenzuela et al. (2021)	1:00	3	85.1	91.2	97.4	70.8	79.4	89.1
Vandewalle et al. (1997)	1:20	4	103.3	101.1	98.6	100.2	99.5	98.1
Means ^b	2:52.5	–	78.7 ± 18.4	85.4 ± 13.0	93.2 ± 7.4	65.3 ± 23.9	72.4 ± 20.8	81.6 ± 16.1
Discrepancies			21.3	14.6	6.8	34.7	27.6	18.4

Values are based on published group means from a 20-study (21-cohort) sample

^aTotal number of employed trials ('All') was equally divided into Short- and Long-trial groups. When 'All' was 3 or 5, the middle trial served in both groups

^bAll differences are highly significant ($p < 0.001$)

et al. 2021; and Pethick et al. 2020, with 3:44-, 4:12-, 5:00-, and 6:12-min shortest trials, respectively). Again, non-linearity is not confined to protocols employing 'exclusion zone' trials.

The bias associated with short trials is moderated by being related to all trials, since the latter also include the former. Comparison with the long trials appears more appropriate because longer trials have been shown to render better MLSS approximations. Reliance on just the long trials, however, is confounded by the extra difficulty of producing true maximal performances in protracted exertions and the resultant lower reliability of such efforts (see 'Trial Duration Effects', above). In endurance-trained cyclists, Abbiss et al. (2007) found up to 8% difference among five 30-km modified time trials. A more extreme example is a 5-trial CP study by Housh et al. (1989), where observed power in the 4th (~33-min long) trial was 9.4% short of that predicted by the preceding 3 trials, while the 5th trial (~57 min) was 16.5% less than predicted by its preceding 4 trials. Although

most reported variabilities are typically quite smaller than these examples, it must be realized that even the best of several 'maximal' performances is not necessarily the true absolute maximum.

Thus, it is reasonable to take ~5% under-performance as a realistic estimate for illustrating the potential effect long-trial bias can have on short-term power and CP. Table 2 shows this via a theoretical, perfectly linear example ($W-t$ $r^2 = 1.00$) of a 3-trial CP-test (3, 8, and 15 min at 300.0, 221.9, and 200.0 W, respectively). Assuming a 5% difference between observed and true maximal power output in the 15-min trial (210 vs. 200 W), the changes in predicted 30-, 60-, and 120-s power outputs would be 13.1, 9.1, and 4.0% lower, respectively (also, note a 4.7% CP increase). These approximations then ought to be subtracted from the corresponding long-trial-based values of Table 1 (34.7, 27.6, and 18.4%). Consequently, the net discrepancies would then be 21.6, 18.5, and 14.4%. Allowing for sample limitations

Table 2 The bias expected in the 30-, 60-, and 120-s-predicted power outputs in a 3-trial sample CP test, due to a possible 5% under-performance in the longest trial (see text for more details)

	CP Trials			r^2	Derived values			CP
Trial duration, min (s)	3 (180)	8 (480)	15 (900)		0.5 (30)	1 (60)	2 (120)	
Work								
Observed, kJ	54.0	106.5	180.0	1.000	28.0	33.0	43.0	
‘True’, kJ	54.0	106.5	189.0	0.999	24.3	30.0	41.3	
Power								
Observed, W	300.0	221.9	200.0		933.3	550.0	358.3	175.0
‘True’, W	300.0	221.9	210.0		811.5	499.8	344.0	183.2
‘True’/Observed, %			105.0		86.9	90.9	96.0	104.7
Discrepancy, %			5.0		– 13.1	– 9.1	– 4.0	4.7
21-sample long-trial discrepancy (Table 1), %					34.7	27.6	18.4	
Approximate net discrepancy, %					21.6	18.5	14.4	

‘True’ = Assumed true maximal performance capacity

and variability, realistic discrepancies of ~25–10% need to be accounted for in the 30- to 120-s duration range.

The seeming linearity of the W – t and P – $1/t$ relationships has prompted researchers to regard linear regression as the best approximation in applying the W – t and P – $1/t$ models. The presented evidence, however, demonstrates that by and large, and even when shortest trials are well within the duration consensus, linear regression is not the best representation of the data. The inescapable conclusion is that observed CP (or CS, CT) discrepancies are not due to random experimental errors but rather to forcing linear models on inherently non-linear data, whether in the P – $1/t$ or the W – t models. A great deal of work has gone into investigating and discussing the different CP/CS determination ‘models’ and their respective merits and drawbacks (Bergstrom et al. 2014; Clark and Macdermid 2021; Leo et al. 2022; Mattioni Maturana et al. 2018; Morton 1996; Sawyer et al. 2014; Sreedhara et al. 2019), but the presumption of linearity has precluded attempts to elucidate the reason behind the discrepancies. The issue, therefore, has consequently never been resolved.

Why are higher power outputs more energetically compromised?

In an article unrelated to CP, Williams (1985) pointed out the interactive bio-mechanical and metabolic complexities of human movement and its efficiency, concluding that “The lack of definitive information concerning the relationship between mechanical and physiological energy changes limits the accuracy, meaningfulness, and usefulness of measures of mechanical power and muscular efficiency”. In direct reference to CP, Vandewalle et al. found it important to stipulate in their remarkably insightful review (Vandewalle et al. (1997), that the CP concept must “assume that performance is determined by metabolic factors” and that “it is also implicitly assumed that mechanical efficiency or energy

cost are *independent* of power or velocity”. This was echoed by Morton (2006) who stressed the requisite presumption that “The efficiency of transformation of metabolic energy to mechanical energy is *constant* across the whole power (and time) domain(s)”. Thus, had mechanical power outputs always constituted a fixed fraction of the corresponding metabolic cost (i.e., fixed efficiency), and as long as CP, like MLSS or MMSS, is regarded as physiological/metabolic rather than a mechanical entity, then there would be no discrepancies between theoretical and observed power outputs. That is, power outputs from shorter or longer trials would yield similar, if not identical CP values.

In their insightful review, Vandewalle et al. (1997) recognized that increasingly higher mechanical power outputs constitute progressively decreasing fractions of their expected values and the associated metabolic costs. In referring to running, they suggested that “the energy cost is probably higher for the running exercises corresponding to the smallest values of t_{lim} and D_{lim} ..., which should induce an anti-clockwise rotation of the t_{lim} – D_{lim} relationship with an overestimation of critical velocity...”. The ‘anti-clockwise rotation’ is the counter-clockwise shift that can be seen in Fig. 3, where the depressed power outputs of the shorter trials cause an elevation of the power asymptote, i.e., CP. Regrettably, this keen observation has not been further pursued.

The discrepancies between observed and expected power output may never be completely resolved in a practical and precise manner. Nevertheless, several factors discussed below can wholly or partly account for the observed deficiency and compromised efficiency of shorter-duration power outputs, as alluded to by Vandewalle et al. (1997).

Metabolic cost of high-frequency contractions

Muscle activation energy Muscle activation involves energy expenditure that presents as heat production which

is not due to the amount of ATP used or the muscle's power output. This has been termed “activation heat” and ascribed to Ca ion release from the sarcoplasmic reticulum and its subsequent rebinding, as well as to “unexplained enthalpy” possibly reflecting conformational changes of the contractile proteins (e.g., Yamada 2017).

Chasiotis et al. (1987) and Bergström and Hultman (1988) compared continuous contractions of the quadriceps muscle to intermittent contractions of increasing frequency in fixed, 1:1, contraction/rest ratios of 3.2/3.2, 1.6/1.6, and 0.8/0.8s, under blood-flow occlusion. This design made for identical total work over the 51-s test duration. Among others, fatigue (end-force), ATP utilization, and $[La^-]$ were measured. Fig. 7A shows how, with increasing contraction frequency, ATP utilization and $[La^-]$ rose in near perfect unison ($r = 0.998$), while end-force precipitously declined (Fig. 7B). As contraction frequency (cadence, stride rate) increases with increasing exercise intensity, the associated activation heat and unproductive energy loss increases as well.

Hogan et al. (1998) conducted a similar study, sadly in dogs (canine gastrocnemius muscle), at two frequencies, identical work rates, and fixed 1:3 contraction/rest ratios of 0.25/0.75 and 1.0/3.0s, for 3-min. Unlike the Chasiotis et al. (1987) and Bergström and Hultman (1988) studies, the muscles were fully perfused (i.e., unobstructed oxidative metabolism). The authors measured numerous variables, most of which are shown in Fig. 8. Although muscle force declined 24% more following the higher-frequency- than in the corresponding low-frequency contractions, its respective ATP utilization, ATP/Force, and VO_2 /Force ratios were 32,

70, and 40% higher, while the corresponding muscle $[La^-]$ and $[H^+]$ were, respectively, 128 and 27% higher. Again, higher-frequency contractions were associated with markedly higher metabolic costs.

Fibre-type-dependent efficiency Increasing exercise intensity is characterized by increasing recruitment of the fast-twitch, type-II motor units. Based on sub-maximal VO_2 measurements, Coyle et al. (1992) showed delta efficiency (ratio of power-output change to change in expended energy) to be highly correlated with type-I muscle-fibre composition ($r = 0.848$). This finding suggests that exercise efficiency decreases with increasing intensity and type-II motor-unit involvement. While this is likely a valid expectation for submaximal trials of ~5 min and longer, the assumption is confounded by the fact that with the increasing cadence and contraction velocities, typical of higher exercise intensities, there is a concomitant rise in type-II motor-unit efficiency and a corresponding decrease in type-I motor-unit efficiency. Sargeant and Rademaker (1996), as well as Sargeant (2007), depicted type-II motor-unit efficiency as peaking at ~140 rpm, and assuming a 50–50% fibre-type distribution, the efficiency ‘break even’ point of the two fibre types as occurring at ~90 rpm. Thus, it is possible that in as much as the two fiber types are concerned, efficiency may decline up to ~90 rpm and then increase at higher cadences. That figure, of course, would change with individual fibre-type composition. Either way, relative motor unit efficiency appears to have negligible or small positive effects on overall power output and efficiency at high-cadence, high-intensity exercise bouts.

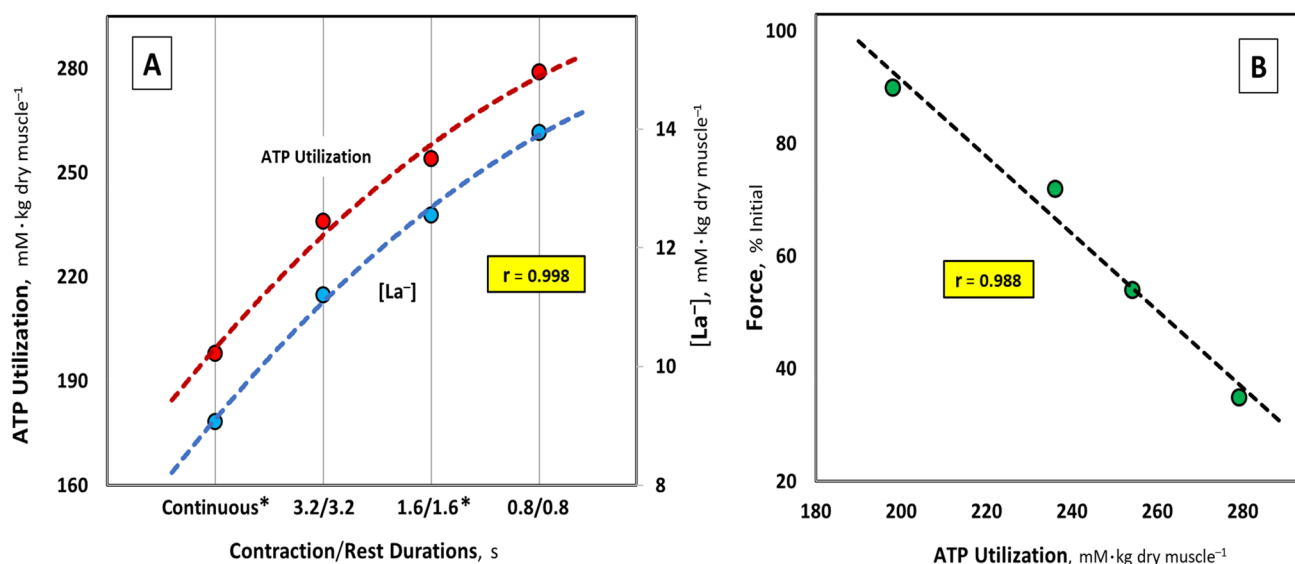
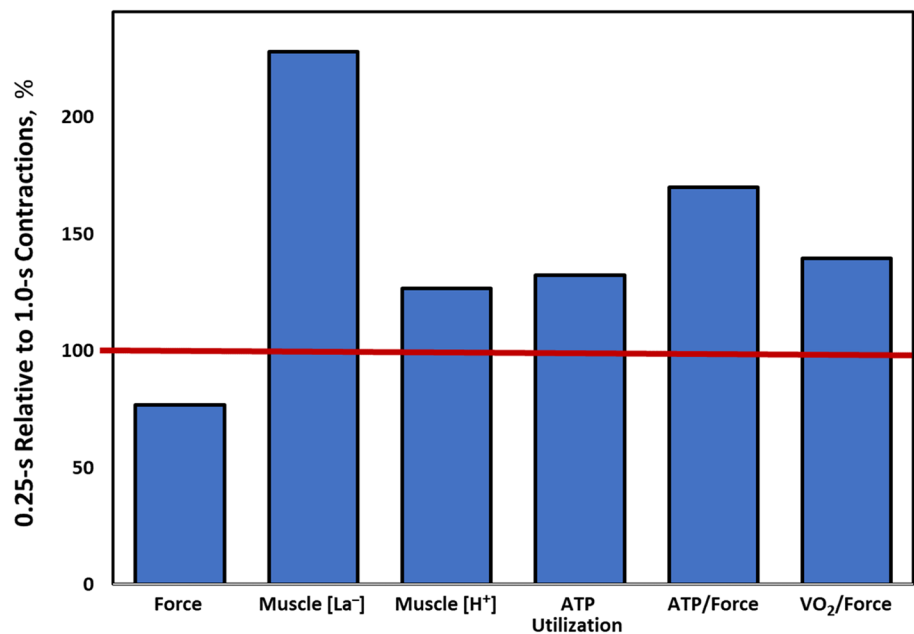


Fig. 7 The effects of different contraction-duration frequencies at identical work rates (1:1 work–rest ratios) on key performance and metabolic variables, based on data from Bergström and Hultman (1988). : ATP utilization and $[La^-]$ in electrically-stimulated knee

extensions, as a function of increasing contraction frequencies. Note the high correlation between the two variables. **B** Decreasing knee extension force despite increasing ATP utilization. *Data from Chasiotis et al. (1987)

Fig. 8 Differences in contractile and metabolic responses to high vs. low intermittent contraction/rest frequencies (0.25/0.75 s vs. 1.0/3.0) carried out for 3 min in isolated canine muscles. The 0.25-s, higher-frequency contraction values are given as percentages relative to the slower, 1-s contractions. Data from Hogan et al. (1998). As in Fig. 7, higher-frequency contractions are associated with greater metabolic cost



Inertial energy losses due to reciprocating leg action

Acceleration/Deceleration of reciprocating leg masses Akin to losses incurred by the reciprocating action of piston engines, leg acceleration and deceleration constitute a major source of energy losses in both cycling and running. These losses might include internal ones, but are mostly due to external losses associated with leg mass acceleration and deceleration, which do not contribute to the measured power output. Had the same fixed cadence been utilized in all studies and at all power outputs, this would not be a differentiating factor. However, since both metabolic- and performance-optimal cadences, as well as self-chosen cadences tend to increase with power output (see under ‘Optimal vs. Selected Cadence’, below), the discrepancy between mechanical output and its metabolic cost grows ever larger with increasing intensity and cadence (see also Table 1).

Decreasing efficiency, or increasing metabolic costs with increasing cadence at given power outputs has been shown numerous times (e.g., Gaesser and Brooks 1975; Widrick et al. 1992). Foss and Hallén (2004) measured both VO₂ and heart rate in unloaded cycling at 60, 80, 100, and 120 rpm and found very large cost increments. While supporting practically no external work (unloaded cycling), heart rate rose 39% while VO₂ increased 102% (0.86 LO₂ min⁻¹, from 0.84 to 1.70) between 80 and 120 rpm (Fig. 9) (80 rpm is taken as reference since it approximates cadences typically used in sub-maximal exercise). Extrapolated to 140 rpm the difference rose to 1.53 LO₂ min⁻¹ (182%). These large differences were likely due, in part, to increasing agonist–antagonist co-contraction stemming from the unloaded nature of the increasing cadence. These differences, therefore,

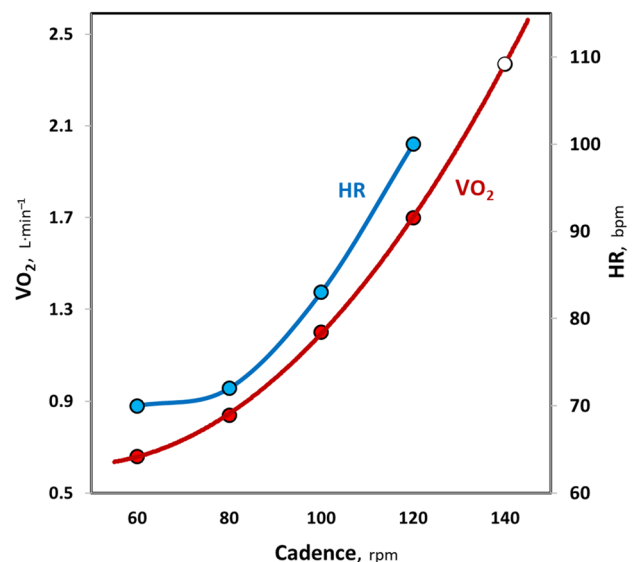


Fig. 9 Increasing heart rate and metabolic cost of unloaded cycling (Foss and Hallén 2004). VO₂ at 140 rpm (open circle) is extrapolated from the data. Relative to 80 rpm, VO₂ increased by 43, 102, and 182% at 100, 120, and 140 rpm, respectively. Heart rate increases from 80 to 100 and 120 rpm were 15 and 39%, respectively

are not expected to increase with power output, but at 4.0 LO₂·min⁻¹, 100, 120, and 140 rpm could still constitute up to ~9, 22, and 38% energy losses, respectively.

Using both loaded and unloaded cycling, Francescato et al. (1995) provided a particularly illustrative depiction of cycling energetics at increasing cadences. Although run on only four young men, the study’s meticulous sophistication warrants close attention to its findings (Fig. 10). VO₂

was measured at 40, 60, 80, and 100 rpm under loaded and unloaded conditions, as well as with and without leg-mass augmentation. Net VO_2 (loaded minus unloaded) shows near-zero values (highest efficiency) at 60 and 80 rpm, but steeply rising metabolic costs at both higher and lower cadences (Fig. 10A). Calculated internal work (mostly mechanical energy due to leg-mass acceleration–deceleration) is different (Fig. 10B). It is practically zero at the two lowest cadences (40 and 60 rpm) but then exponentially grows with rising cadences. It is a modest rise of $0.23 \text{ W}\cdot\text{kg}^{-1}$ from 60 to 80 rpm, but from 80 to 100, 120, and 140 rpm (the latter two are extrapolated), the increases are exponentially higher: 0.75, 2.16, and $4.56 \text{ W}\cdot\text{kg}^{-1}$, respectively. Notably, $2.16 \text{ W}\cdot\text{kg}^{-1}$ is equivalent to $\sim 50\text{--}80\%$ of most peoples' maximal aerobic power, while $4.56 \text{ W}\cdot\text{kg}^{-1}$ is $\sim 80\text{--}100\%$ of maximum of well-trained to elite cyclists.

Increasing energy losses mean decreasing efficiency. Sanderson (1991) tested 38 recreational and 7 competitive cyclists at 60, 80, and 100 rpm. At a 235-W workload, mean efficiency declined 29% between 60 and 100 rpm (Fig. 11), 11% less than the 40% reported by Blake and Wakeling (2015) in a nearly identical cadence range of 60–98 rpm. The efficiency drop upon raising the cadence from 80 to 100 rpm was 18%, while further raising the cadence to the extrapolated 120 and 140 rpm, the efficiency declined 34 and 50%, respectively. Surprisingly, no efficiency differences were detected between cyclists and non-cyclists, although trained cyclists have been shown to have better muscle coordination (e.g., Chapman et al. 2009; see also under 'Muscle Coordination Loss and Agonist–Antagonist Co-Contraction', below).

Such efficiency changes have direct effect on CP determination. For example, at 100 compared with 60 rpm, Hill et al. (1995) reported 4.4–5.8% lower CP (depending on the

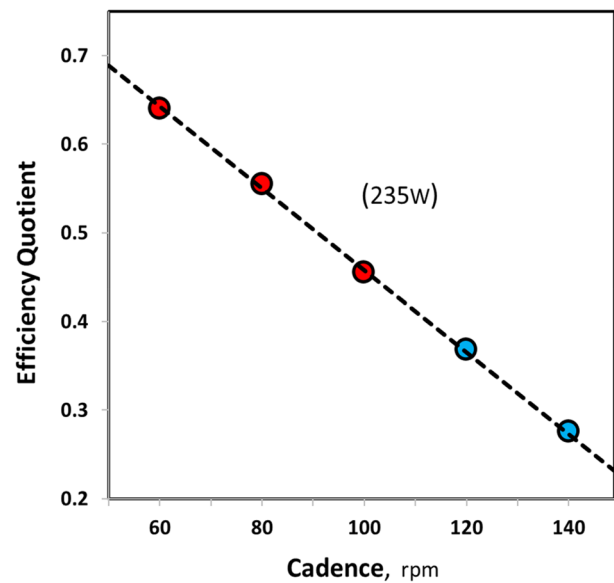
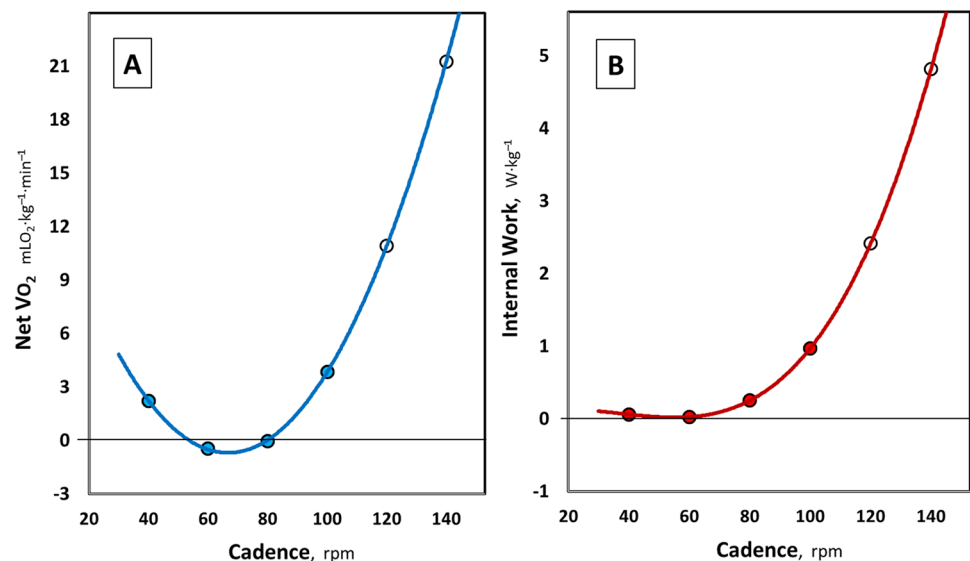


Fig. 11 The dropping efficiency of increasing cadence at a fixed power output (235 W), measured at 60, 80, and 100 rpm in 45 cyclists (Sanderson 1991). The drop in efficiency from 80 to 100 rpm is 18.0%, while to the extrapolated 120 and 140 rpm (blue markers) the respective drops are 33.7 and 50.3%

model used), while Barker et al. (2006) and Dekerle et al. (2014) found 8.7 and 13.5% lower CP, respectively, using the same cadences.

Muscle coordination loss and agonist–antagonist co-contraction Good economy of the cyclical, reciprocating leg movement, in both cycling and running, requires precise muscle coordination, in general, and of agonist–antagonist activation and deactivation, in particular. Both optimal

Fig. 10 Net VO_2 and internal work (energy) losses due to the reciprocating leg action. Data from Francescato et al. (1995). The original data points at 40, 60, 80, 100 rpm (filled circles) were polynomially extrapolated ($r = 1.00$) to 120 and 140 rpm (open circles). **A** Net VO_2 (loaded minus unloaded) vs. rpm. Note near-zero net VO_2 at 60 and 80 rpm but steeply rising O_2 cost at both higher and lower cadences. **B** Calculated internal work (mainly due to leg-mass acceleration–deceleration at increasing cadences) is near zero at the two lowest cadences but grows exponentially greater thereafter



and self-selected cadences increase with increasing power outputs (e.g., Ansley and Cangle 2009; Coast and Welch 1985; Dekerle et al. 2003; MacIntosh et al. 2000; Rudsits et al. 2018). The rising cadence makes muscle coordination increasingly more difficult to achieve and is another source of energy loss, particularly so with non-cyclists. For example, Chapman et al. (2008) tested 10 novice and 9 highly trained cyclists at cadences ranging from 55 to 95 rpm. Throughout the cadence range, the novices exhibited significantly higher co-contraction than their highly trained counterparts, but with increasing cadence, co-contraction levels significantly rose only the novices. While it has not always been demonstrated (e.g., Sanderson 1991), this ought to be expected since high-level cyclists are well-accustomed to the top end of the tested cadence range as well as to higher cadences. Even well-trained cyclists, however, could be expected to demonstrate increased co-contraction at cadences above ~100–110 rpm. Mainly true for higher cadences, Blake and Wakeling (2015) showed muscle coordination to be a key determinant of mechanical efficiency and power output of limb movement at the high end of the cadence spectrum, largely due to the finite minimal durations of muscle activation–deactivation cycles.

In another study (Blake and Wakeling 2014), the authors related differences in minimal activation–deactivation cycle durations to muscle composition, in line with what has previously been shown by Sargeant (2007) and Kohler and Bou-tellier (2005). That is, persons with dominant slow-twitch (type-I) fibre composition would be expected to have markedly lower efficiency- and power-optimized cadences than would high-type-II individuals.

Optimal vs. Selected cadence

The relationship between cadence and power output is partly clouded by the fact that older studies used friction-based cycle-ergometers, where a given power output had to be maintained by an imposed fixed cadence. However, even since freely chosen cadences have become possible with the advent of electromagnetically-braked ergometers, large differences between optimal vs. self-selected cadences have typically been encountered. In 8 trained cyclists, Marsh and Martin (1993), for example, reported ~60 rpm as energetically most efficient, but mean self-selected cadence was ~90 rpm. Further confounding the issue is large inter-individual differences. For example, mean preferred cadence of 7 competitive cyclists was 91 rpm, but individual values ranged between 72 and 102 rpm (Hagberg et al.'s 1981). A yet wider range of 70–105-rpm was reported by Mattioni Maturana et al. (2016). These and other factors are presumably manifested by the large variance among 15 different studies of submaximal power outputs (<100% $\dot{V}O_2\text{max}$), reviewed by Ansley and Cangle (2009) (Fig. 12). The

cadence–power correlation was only moderate ($r = 0.65$) but highly significant nevertheless ($p < 0.005$). It was also shown that the upward shift in optimal-cadence with rising workload $\dot{V}O_2$, was closely paralleled by a similar increase in optimal EMG_{RMS} , an effect also shown by MacIntosh et al. 2000.

Figure 13 is a composite depiction of optimal cadences for the entire spectrum of power outputs, including supra-maximal efforts close to 2000 W (~22 $\text{W}\cdot\text{kg}^{-1}$), based on the data of 6 different studies (Blake and Wakeling 2015; Emanuele et Denoth 2012; Gardner et al. 2003; Kordi et al. 2019; MacIntosh et al. 2000; McCartney et al. 1985; Rudsits et al. 2018). It demonstrates that cadences higher than 100 rpm and as high as 140 rpm or more may be required to maximize short-term power. It ought to be pointed out that the instantaneous peak-power cadence of 160 rpm is from match-sprint track cyclists using fixed gears. Gear selection in track cycling is an optimization compromise between acceleration and power, i.e., not optimized for instantaneous peak power. For this reason, this data point was not included in the regression. It is highly likely that true optimal cadence for those cyclists was closer to 140 than to 160 rpm with even a higher peak power to boot. To summarize, selecting wrong resistance loads or otherwise forcing non-optimal cadences will inevitably result in lower power outputs than would otherwise be possible.

An insight into the issue of cadence-related energy losses is brought to light by the discrepancy between AV Hill's classical hyperbolic force–velocity relationship (Hill 1938) and cycling's linear pedal-torque–cadence relationship (Douglas et al. 2021; Driss and Vandewalle 2013; Jaskólska et al. 1999; Rudsits et al. 2018; Sargeant et al. 1981;

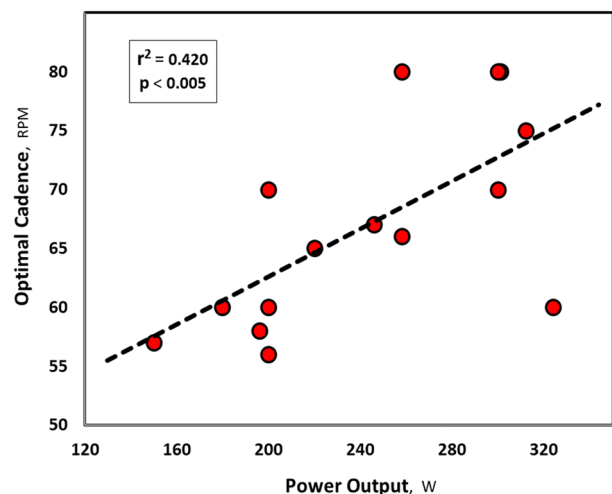


Fig. 12 Optimal cadence vs. power output relationship—a composite of 15 different studies reviewed by Ansley and Cangle (2009). Despite a significant degree of dispersion, due to various factors (see text), the relationship is statistically highly significant

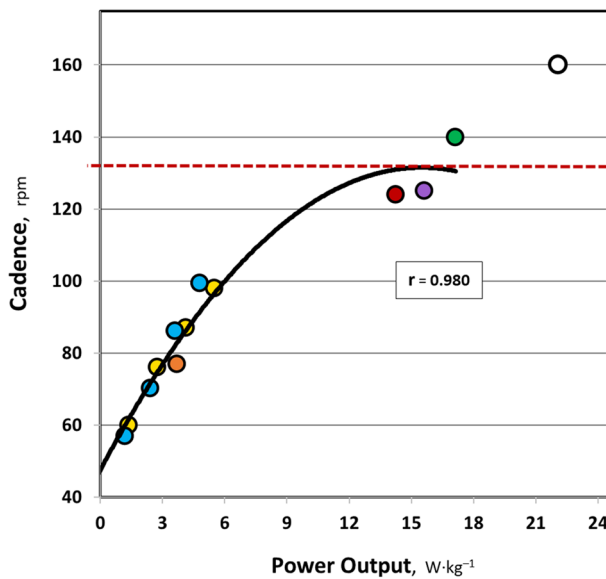


Fig. 13 Optimal cadence as a function of body-mass-normalized power output ($\text{W}\cdot\text{kg}^{-1}$). Data are from Blake and Wakeling (2015) (yellow), Emanuele and Denoth (2011), Gardner et al. (2003)* (white), Kordi et al. (2019) (red), MacIntosh et al. (2000) (turquoise), McCartney et al. (1985) (green), and Rudsits et al. (2018) (purple). Regression peak is at 132 rpm, but data points are group means and individual optimal cadences can vary widely. For example, McCartney et al.'s cyclists averaged 140 rpm, but individual optima ranged between 120 and 160 rpm. *Match-sprint track cyclists. Not included in the polynomial regression (see text)

Vandewalle et al. 1997). This is schematically depicted in Fig. 14. The discrepant gap between the two functions (gray area) is largely due to the increasing energy losses incurred with rising cadences. This prevents the muscles from attaining higher contraction velocities that could maximize power output. Quasi-linearity of the force–velocity relationship, partly ascribed to segmental acceleration, was also described in a multi-segmental leg-press model (akin to cycling's power stroke), but the prime mover (m. vastus lateralis) still exhibited the classical hyperbolic relationship (Bobbert 2012). The segmental model, however, differs from cycling in that the leg's movement is linear rather than circular, there is no deceleration and consequently also no acceleration–deceleration transitions. Regardless, the very fact that cadence cannot follow the hyperbolic velocity increase of freely contracting muscle is a major factor in limiting mechanical power output.

Thus, power outputs at all cadences high enough for relevance in maximal cycling or running performances, are limited by energy losses that increase with increasing cadences. Included under these losses, aside from the internal work associated with limb-mass acceleration and deceleration, is the frequency-related cost of activation, and diminishing agonist–antagonist coordination.

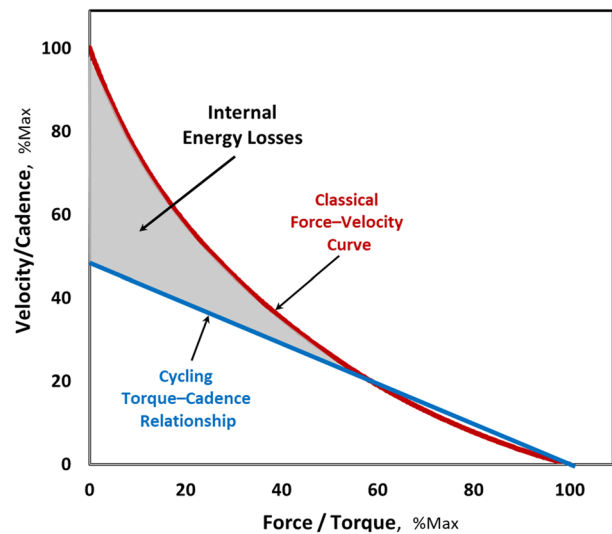


Fig. 14 Schematic depiction of the discrepancy between the linear cadence–torque relationship in cycling (e.g., Vandewalle et al. 1987; Driss and Vandewalle 2013) and the classic hyperbolic force–velocity relationship (Hill 1938). The gap between the two (at lower forces/torques and higher velocities/cadences) is largely due to the increasing internal energy losses with increasing cadences in cycling

Cost of increasing torques

At submaximal aerobic exercise intensities and at fixed cadences, Gaesser and Brooks (1975) and Widrick et al. (1992) found cycling efficiency to increase with increasing power outputs. Maximal aerobic power was not measured, but the tested maximal intensities were quite moderate (~130 and 146 W, respectively). Blake et al. (2012), also studied submaximal exercise, but to much higher intensities (90% VO_2max). They found the highest mechanical efficiencies to occur at 55–60% VO_2max and deteriorate thereafter. These relative intensities roughly correspond to the 130–146 W of the aforementioned studies and there is thus no conflict between findings. Blake et al. went on to state that there is “a trade-off between power and efficiency in cycling because the highest mechanical efficiencies did not occur at the highest power outputs”.

The decreasing efficiency above 55–60% VO_2max should not be confused with that associated with increasing cadence because the cited power-output findings refer to fixed cadences. Hence, this appears to be a different case of energy losses. In four untrained young men, Kaneko and Yamazaki (1978) calculated energy losses in relation to rising pedal loads, by increasing flywheel frictional loads from 1 to 5 kp (kg force), applied at a fixed ~52-rpm cadence

(Fig. 15). The lowest internal work occurred at the 2-kp load, roughly corresponding to 50% $\dot{V}O_{2\max}$ in untrained individuals and was not statistically different from the 1-kp load. The last three loads (3, 4, 5kp), however, markedly increased internal-work (energy losses) by 3.7, 21.0, and 29.7%, respectively. The authors did not provide maximal aerobic power (MAP) data, but the MAP of Vandewalle et al.'s (1997) nine untrained participants, for example, was $3.56 \text{ W}\cdot\text{kg}^{-1}$. Using this as reference, the 4th and 5th loads (3.24 and $4.05 \text{ W}\cdot\text{kg}^{-1}$) would have constituted 91 and 114% MAP, respectively—well inside the severe-exercise domain. It is illuminating to compare these findings to those depicted in Fig. 10A, where the net $\dot{V}O_2$ increase from 60 to 40 rpm can be ascribed to the 50% increase in the pedal load necessary for maintaining the given power output.

Thus, load-related energy losses significantly increase with increasing intensities up to 100% $\dot{V}O_{2\max}$ and there is no reason to assume they do not continue this trend at supra-maximal intensities. However, standard efficiency measurements ($\dot{V}O_2$) do not apply at those intensities and are impossible or difficult to quantify. At any rate, this decreased efficiency does compromise measured high-intensity/short-duration power outputs.

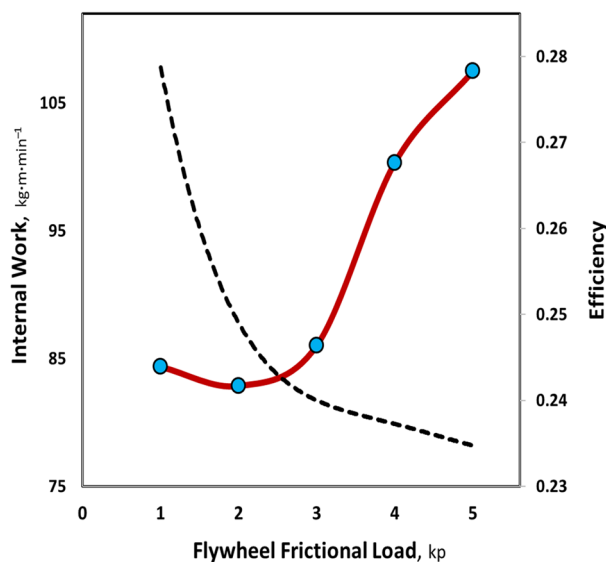


Fig. 15 The rise in internal work with increasing pedal load (flywheel frictional load) at a fixed cadence of ~52 rpm (red line) and the concomitant decline in cycling efficiency (dashed line). Data from Kaneko and Yamazaki (1978). In parentheses, the associated power outputs in $\text{W}\cdot\text{kg}^{-1}$. Note the sharply increasing internal work at the last two loads that are in the severe exercise domain

Which optimum?

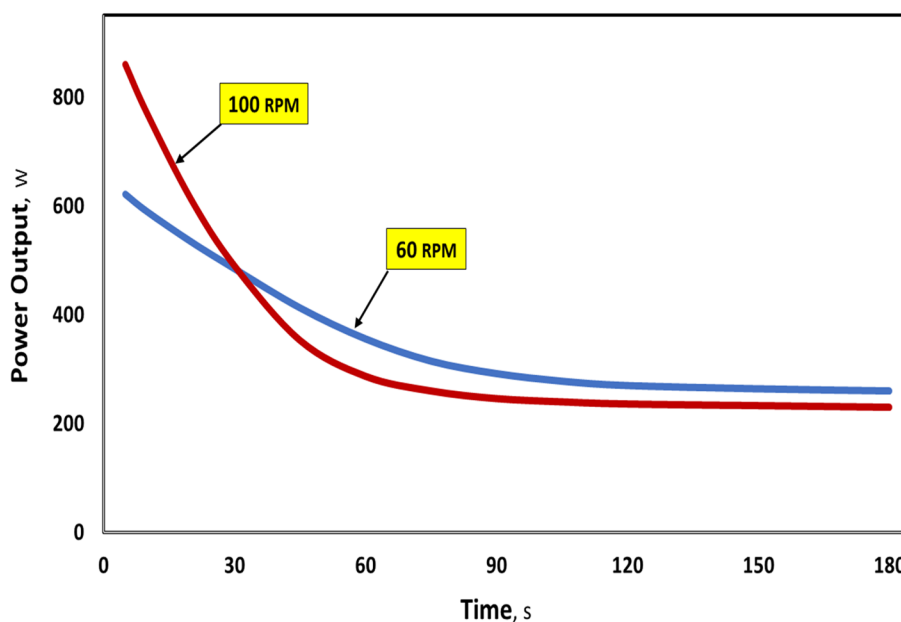
Although it may superficially seem a paradox, higher power outputs are generally associated with higher cadences, including those at MLSS (e.g., Beneke and Leithäuser 2017), while cycling economy decreases with rising cadence (above ~55–60% $\dot{V}O_{2\max}$). Kohler and Boutellier (2005) addressed this issue, showing that the optimal cadence for maximizing power output is 1.68- to 2.44-fold higher than the energetically most efficient cadence, a ratio that depends on the prevalence of type-I (lower values and ratios) vs. type-II muscle-fibres (higher values and ratios). Dekerle et al. (2014), for example, administered the 3MT at 60 and 100 rpm (Fig. 16). The changing relationships demonstrate 100 rpm as generating superior power outputs up to ~ 30s into the 180-s effort, as would be expected based on the characteristics of the force–velocity relationship (Hill 1938; Driss and Vandewalle 2013). From ~ 30s on, however, 60 rpm was more advantageous, resulting in higher end-power (259 vs. 227 W) and CP values (245 vs. 212 W).

Thus, metabolically optimal and performance-optimal cadences are never the same and one is left with the question of whether to prefer efficiency while sacrificing maximal performance, or vice versa. There apparently is no way to maximize both, which is one of the main reasons why observed outputs are doomed to underestimate metabolic costs.

Overview

As alluded to earlier, the underlying cause of the rising metabolic costs of increasing cadence, as depicted in Figs. 9 and 10, is not unifactorial but rather an effect sum of several factors (acceleration/deceleration, muscle activation, coordination/co-contraction). To this can be added the cost of increasing pedal load. Taken at face value, the combined effects appear sufficiently large to account for the observed–expected power output discrepancies as summarized in Tables 1 and 2. Nevertheless, it is conceivable that the discussed-factor list is not comprehensive and that other factors might be at play to one extent or another. On the other hand, based on the internal work values of Fig. 10B, it is hard to see how it is humanly possible to achieve maximal power outputs at 160 or even ‘only’ 140 rpm (Fig. 13). However, since Fig. 10 values are based on a limited sample of untrained individuals, it is possible that they constitute a degree of overestimation of true population means. While that can certainly be true, a more potent explanation is that athletes or other individuals reaching those cadences have significantly better coordination, less agonist-antagonist co-contraction, and significantly higher type-II muscle fibre composition than those on whom Fig. 10 is based.

Fig. 16 Schematic depiction of the power output vs. time relationship in a 3-min all-out cycling test, based on Dekerle et al. (2014). Note that the 100-rpm power output was superior to that of 60 rpm for the first ~30 s. From ~30 to 180 s, however, the lower efficiency (greater internal-energy losses) associated with the higher cadence brought its power output to below that of the lower, more efficient cadence



Exceptions?

There are a few exceptions to the overwhelming general observation of CP/CS > MLSS. Some may be readily explicable while some might not, possibly due to insufficient data. Here are four examples where CP or CS appear to closely approximate MLSS.

Wakayoshi et al. (1993) determined swimming-flume CS in elite swimmers, using only two reference points of 200- and 400-m all-out swims (respective mean durations: 2:03 and 4:22 min). A greatly modified MLSS testing followed, consisting of 4x400m swims at 98, 100, and 102% CS, with 30–45-s blood sampling intervals. They found the $[La^-]$ response to the 100% CS interval session to be the closest to the modified-MLSS intensity (i.e., $MLSS \approx CS$) and suggested “that V_{crit} [CS] can assess the physical performance without the need for either blood sampling or expensive equipment”. Due to the water’s drag in swimming, power requirement is proportional to velocity cubed and thus, a 2% velocity increase requires > 6% power increase, which means a relatively low test resolution (~ 16 s in 400 m). The study’s main problem, however, was the fact that interval sessions markedly increase sustainable velocities and have thus biased MLSS upwards. Moreover, the 4 × 400 m-set’s total exercise time at 100% CS was <17.5 min (rather than 30 min), thus further overestimating the ‘sustainable’ MLSS velocity. Due to this protocol’s peculiarities MLSS velocity was, therefore, inadvertently brought on par with CS, resulting in a false conclusion.

The Pallarés et al. (2020) study (discussed earlier under ‘Pseudo Linearity’) found CP to be practically identical to MLSS (indeed, 0.4% lower). The study is an excellent

example of the subduing effect long trials have on CP. The 74-min 4th trial effectively ‘pinned down’ the CP to MLSS or lower power output, regardless of any preceding trials, thus making it indistinguishable from MLSS. This raises the fundamental question of why the demanding conventional CP protocol is necessary in the first place when a single long trial is all that is required for a sound CP estimation (see also under ‘Trial-Duration Effects’, above, and ‘Improving Accuracy’, below).

An intriguing study is Vandewalle et al. (1997). Nine untrained men performed the 30-s Wingate Anaerobic Test, an MAP test, and 4 CP trials at 105, 120, 135, and 150 %MAP. There were several exceptional aspects. First, highly linear $P-1/t$ and $W-t$ relationships ($r = 0.9995$ and 0.9998 , respectively) to the extent that the regressions of the two shortest (80, 98s) and the two longest (133, 193 s) trials were practically identical (see also Table 1). Second, all 4 trials were particularly short ($\leq 3:13$ min) with only 1:53 min separating the shortest from the longest. Third, the Wingate test’s power output ($9.77 \text{ W} \cdot \text{kg}^{-1}$) was only 4.2% lower than the $10.19 \text{ W} \cdot \text{kg}^{-1}$ predicted by the $P-t$ hyperbolic relationship, < 20% of the mean expected difference (Tables 1 and 2). This may well be due to the close proximity of the shortest trial’s duration to that of the Wingate test. The near-perfect $P-1/t$ and $W-t$ linearities, on the other hand, may be due to the fact that all trials were in the high glycolytic zone, possibly placing them on a locally near-perfect linearity that would presumably be broken once longer trials were added.

A less explicable example is the apparently well-executed study by Kranenburg and Smith (1996). Nine highly trained male runners underwent two CS tests, once on a treadmill ($CS = 299.7 \text{ m} \cdot \text{min}^{-1}$) and once via track running ($CS =$

292.1 m·min⁻¹; $r = 0.90$). The higher treadmill CS was reasonably ascribed to the lack of wind resistance in stationary running. The runners then competed in a flat, 9.8-km cross-country race. Their race velocities were highly correlated ($r = 0.92$) with both CS determinations and, importantly, mean race velocity was practically identical to the track-based CS. As mean race time was 33.5 min, race velocities must have closely approximated MLSS speeds. It was, therefore, a unique case in which CS not only was exceptionally close to MLSS but could also be sustained for well over 30 min. The published data, however, appear to be erroneous. Reported distances and TTEs for the CS determination trials render velocities that are higher for the middle and long trials than for the shortest one. It is thus impossible to further evaluate the otherwise interesting findings, which is also why the study was not included in Table 1.

Critical torque

Like CP and CS, CT is afflicted by overestimation of the sustainable effort (e.g., Abdalla et al. 2018; Hendrix et al. 2009; Kellawan and Tschakovsky 2014). CP and CT are similar in behaviour but due to their different natures (power vs. torque) they are not entirely comparable.

Parallels with CP

CT is also assumed to be the asymptotic torque of a hyperbolic torque–time relationship. As in CP, this assumption is usually not realized, also due to the non-linearity of the W – t and Tq – $1/t$ relationships. Indeed, the ‘exclusion zone’, where this deviation from linearity is most pronounced, was first defined, as 5 min (2 min in intermittent contractions), by Monod and Scherrer (1965) in isometric force testing of discrete muscles or muscle groups.

As with CP, the underlying reason for non-linearity and the introduction of the ‘exclusion zone’, is the lower-than-expected torque values at the short end of the duration continuum. Since type-II motor units become increasingly recruited with increasing contraction intensity, their relative efficiency, if different from that of type-I, can become a differentiating factor in determining the observed-vs.-expected discrepancy. Unlike in high-cadence or other high-velocity contractions, where type-II motor units become increasingly more efficient (see above, under ‘Fibre-Type-Depended Efficiency’), the situation is reversed under isometric conditions. An excellent demonstration of that are the findings of Han et al. (2001), who measured tension cost (ATP cost per unit isometric tension) in myosin heavy-chain (MHC) isoforms of human ($n = 12$) vastus lateralis. The tension costs of the MHC_{2A} and MHC_{2X/2A} (main constituent isoforms of type-II muscle fibres) were 2.75 and 3.35 times greater than that

of the MHC_{slow} isoform (chief constituent of type-I muscle fibres), respectively. He et al. (2000) reported similar findings in isolated whole fibres of the four main types. In practice, differences are not as extreme since muscles are not uniformly made up of a single fibre types and a given muscle fibre is not homogeneously made up of just one or two MHC isoforms (e.g., Ennion et al. 1995).

Partial and complete blood occlusion

While energy losses, due to reciprocating limb action, are of no relevance to CT, another factor, inconsequential in CP determination, is at play with CT. Blood-flow obstruction can range from complete patency in low-intensity/long-duration isometric contractions, to complete occlusion at higher intensities. For example, Barnes (1980) tested 20 young men and found complete blood-flow occlusion in the forearm muscles, already at 63.5% of handgrip MVC. However, in the 10 highest-grip-strength participants occlusion occurred already at 51.5%, while only at 75.5%MVC in the lowest-strength group. A significant negative correlation ($r = -0.58$; $p < 0.01$) was found between maximum isometric strength and the MVC fraction necessary to produce intramuscular occlusion. No difference could be shown between the absolute occlusion-producing tensions of the high- vs. low-strength sub-groups (34.8 vs. 34.3 kg, respectively), suggesting that absolute tension (intra-muscular pressure), rather than any given MVC fraction, is what determines the occlusion threshold. Thus, Torque determinations are likely to be skewed downwards (and CP upwards) more by high- vs. low-torque trials and in high- vs. low-strength individuals.

Figure 17 schematically depicts Barnes’ group means of muscle blood flow vs. contraction intensity. Each participant’s blood-flow values were normalized to his resting value. It is noteworthy that blood flow peaked already at ~20%MVC and declined precipitously thereafter. As compromised O₂ supply is attendant to diminishing muscle blood flow, torque performance may start declining already at rather low MVC fractions. It may, therefore, be quite illustrative to note that contraction intensity at the 5-min point of Monod and Scherrer’s (1965) ‘exclusion zone’ was ~26%MVC, marked by the dashed red line. According to this then, for contractile torques to be unaffected by restricted blood flows, durations of ~5 min or longer are required. While this constitutes emphatic support to Monod and Scherrer’s observations, avoidance of the 5-min zone also means complete elimination of the entire ascending arm of the torque–time hyperbolic relationship.

More extreme findings were reported by de Ruiter et al. (2007). In 15 male non-athletes, deoxygenation/reoxygenation of the knee extensors (m. VL, ML, RF) was measured (via near infra-red spectroscopy) as a function of contraction

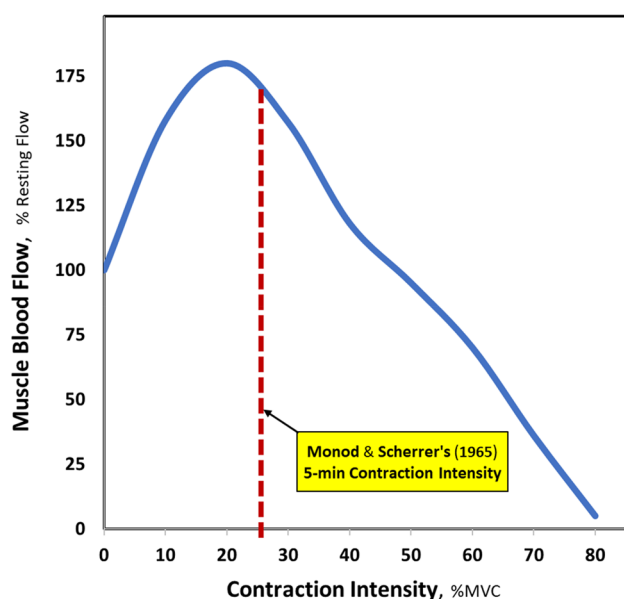


Fig. 17 Muscle blood flow vs. continuous isometric contraction intensity (hand grip), depicted from Barnes' (1980) data. Blood-flow (normalized to each participant's resting value) peaks already at ~ 20%MVC and declines steadily thereafter. The resulting compromised O_2 supply likely explains Monod and Scherrer's (1965) 'exclusion zone' for their continuous isometric contractions, where the force level at the 5-min point was ~ 26%MVC (dashed red line). Shorter contraction durations would be of greater intensity and increasing performance handicap due to compromised O_2 supply

intensity (percentage of maximal stimulated torque capacity; MTC). Ceasing of reoxygenation was taken to indicate complete blood-flow occlusion. Contrary to Barnes' (1980) findings (63.5%MVC) and others', complete occlusion took place already at 24.3%MTC in both vastus muscles and at 33.9% in rectus femoris (since mean volitional activation was 95%, these percentages would be only ~5% higher when expressed as %MVC). However, contrary to Barnes (1980) who found identical occluding torques in his weak and strong sub-groups, de Ruiter et al. reported the occluding torques to be strongly correlated with absolute MTC ($r = 0.72$, $p = 0.003$) rather than being similar. The reason for this discrepancy is unclear but may be related to the different natures of the muscles used in the two studies.

'Anaerobic' work capacity and CP testing

One of the arguments in favour of CP testing, in preference to MLSS or other MMSS estimates, has been the added benefit of 'anaerobic'-work-capacity estimation. In CP terms this is W' (work done above CP). Several studies pointed out an inverse relationship between CP and W' (Housh et al. 1991; Clark and Macdermid 2021; Dekerle et al. 2014). This is to be expected as high-level aerobic endurance is often

associated with low glycolytic capacity and vice versa. The question, however, is how reliable W' is and how representative it is of non-oxidative work capacity.

The Maximal Accumulated Oxygen Deficit method (MAOD) is widely regarded as the method of choice for assessing maximal non-oxidative work capacity, although it has never been standardized, has been criticized on several methodological fronts (e.g., Noordhof et al. 2010), and is not free of conceptual shortcomings. For example, MAOD cannot tease out the metabolic augmentation, due to the exercise-induced rise in body and muscle temperature (Q_{10} effect), from the post-exercise oxygen consumption. Nevertheless, no better method is available and MAOD will be used as a reference for 'anaerobic' work capacity.

Low W' sensitivity

W' variability

Chidnok et al. (2013) tested W' under 4 pacing regimens (incremental ramp, all-out, self-paced, and CWR) in the 3MT. The mean derived W' values differed from each other by as much as 8.5%, and the coefficient of variation between any two protocols was roughly double that (12–19%). Bergstrom et al. (2014) reported even greater variability. Testing five models of CP and W' estimation, the authors found as much as 13.8% CP-estimation difference among the models, and importantly, also >3-fold greater (46%) difference in the corresponding W' values.

Training response

W' 's sensitivity to training has also come under scrutiny, which questions its validity as a measure of non-oxidative work capacity. Jenkins and Quigley (1993) trained 8 previously untrained male students for 8 weeks via high-intensity interval training and, as expected, found a much greater increase in W' (49%) than in CP (3.4%). On the other hand, in an earlier study by the same authors (Jenkins and Quigley 1992), 12 previously untrained male students aerobically trained for 8 wks. CP demonstrated exceptional improvement of 30% (196 to 255 W). W' was not expected to markedly change because of the endurance-oriented nature of the training regimen, but it nevertheless precipitously and inexplicably dropped by 26% (from 19.9 to 14.7 kJ). Due to its very high variability, the drop in W' did not quite reach statistical significance, but a corresponding 23% drop was also observed in the control group that has undergone no training at all.

Kendall et al. 2009 trained 42 recreationally active young men via high-intensity interval training, with and without creatine supplementation, meant to enhance glycolytic capacity. Although CP improved 6.7% with creatine and

3.6% without it, W' showed no improvement whatsoever (MAOD was not tested).

MAOD, on the other hand, has typically been found highly responsive to training, as ought to be expected. Weber and Schneider (2002), for example, reported 22.6 and 19.3% MAOD increase in men and women, respectively, following 8 weeks of high-intensity interval training. Not only glycolytic capacity was improved. Endurance times at the initial MAOD intensity (120% $\dot{V}O_{2\max}$) increased even more: by 73 and 56%, in the men and women, respectively. Following a rather short but more aerobic 6-week training period, Ramsbottom et al. (2001) reported 18.6 and 20.9% MAOD increase in men and women, respectively, while the corresponding endurance-time improvements were 41.2 and 42.6%.

W' vs. MAOD

An apparent paradox is the fact that W' changes opposite to MAOD. Muniz-Pumares et al. (2017) compared MAOD with W' in the all-out 3MT and in a corresponding CWR test. W' was 19% higher in the 3MT than in the CWR test (11.37 vs. 9.55 kJ), but the corresponding value for MAOD was 12% lower (3.68 vs. 4.18 LO_2). This discrepancy, and the fact that the W' –MAOD correlation could explain only 43% (CWR) and 44% (3MT) of the observed W' –MAOD variance, prompted the authors to conclude that the two “measures have different underpinning mechanism”. In effect saying that W' is not a proper substitute for MAOD.

Why is W' different from MAOD?

Different definitions

A fundamental difference between W' and MAOD lies in how they are defined. While W' accounts only for work done above CP, MAOD includes the O_2 deficit incurred before $\dot{V}O_2$ stabilizes and reaches CP level. This is one of two major reasons why W' is invariably found to be smaller than MAOD.

Mechanical vs. Metabolic energy

The other major reason for the W' –MAOD difference lies in what is being measured. While W' is derived from mechanical power outputs, MAOD is determined metabolically by measuring actual O_2 consumption. The typical MAOD testing duration of ~ 2–3 min coincides with the region of greatest discrepancy between the measured (mechanical) power output and that expected based on metabolic cost. Since W' is derived from those compromised mechanical power outputs, its magnitude is compromised as well. Using Muniz-Pumares et al.'s (2017) data, a numerical comparison will show this more succinctly. Mean MAOD (CWR and

3MT) was 3.95 LO_2 . Assuming metabolic efficiency of just 20%, that value is equivalent to 16.53 kJ, while the corresponding mean W' was only 10.46 kJ. That is, a W' /MAOD ratio of only 0.63. With metabolic efficiency any higher than 20%, MAOD would be higher and the ratio even lower.

Cadence effect

Suboptimal cadence lowers the power output, thus directly decreasing the calculated W' . Dekerle et al. (2014), for example, reported W' at 100-rpm (20.6 kJ) to be 27% higher than that at 60 rpm (16.2 kJ). This W' difference was inversely-related and twice the magnitude of the corresponding CP difference. Using the same two cadences (60 and 100 rpm), similar results were also reported by Barker et al. (2006).

Thus, while the MAOD method is not beyond reproach, W' is not a veritable MAOD substitute and cannot be used as a justification for preferring CP testing over other means of MMSS / heavy–severe exercise boundary assessment.

CP Testing vs. Alternatives

Inadequate resolution

Although CP proponents stand by its validity, most researchers agree that it is inaccurate in predicting endurance performance. Depending on the cadence being used (e.g., Dekerle et al. 2014), the number and duration of trials, or models employed (Bergstrom et al. 2014; Bishop et al. 1998; Clark and Macdermid 2021; Malaguti et al. 2006; Mattioni Maturana et al. 2018), CP determinations often vary by 6 to > 20%. Such high variability and low resolution are incongruent with the level of accuracy required for usefulness by athletes and coaches. For example, just a 4% error in a 40-km cycling time-trial, covered in ~50 min, would be ~ 2 min, which is more than twice as large as what the coach and rider would likely already know, based on training and racing data.

Thus, at least in its current common form, CP testing is not suitable for athlete evaluation. Furthermore, as a multi-trial, time-consuming test that also requires an exceptionally high level of motivation, CP determination is highly impractical for anybody—particularly so for the general population and mass testing, where a basic, non-invasive exercise tolerance, or lactate response tests would be more time-effective and at least as informative.

Improving accuracy

Protocol standardization (e.g., trial spacing and durations, pacing regimen, and model being used) could go a long way

toward reducing CP's variability and increasing its accuracy in approximating MLSS/MMSS. Long trials have been shown to do just that. The 35-min trials used by Vautier et al. (1995), and the recommended 35 and 40 min ones (Vandewalle et al. 1997; Hill 1993, respectively), constitute excellent practical advice. The 35- and 40-min durations again raise the question of why CP testing is necessary (see earlier discussion under 'Exceptions?'). TTEs at MLSS intensity are over 30 min and up to nearly twice that duration, e.g., 34.4 and 37.7 min (runners and cyclists; Fontana et al. 2009), 44 min (runners; Dittrich et al. 2013), and in the well-trained even 55 min (cyclists; Baron et al. 2008). Thus, 35- to 40-min maximal efforts are already exceedingly similar to MLSS intensities. As such, they likely are better MMSS predictors on their own than any or most combinations of 2–5 shorter efforts (see again Pallarés et al. 2020).

Hyperbolic non-compliance

This review has demonstrated that while the elegant hyperbolic P - t relationship is likely conceptually valid, there is presently no way by which to correct the distortion caused by the changing ratios between mechanical power and its true metabolic cost. This and the noted shortcomings of W' determination render the application of the CP concept inferior to MMSS estimation by alternative present-day tools.

Testing alternatives

MMSS determination (or the heavy–severe exercise boundary) is not a case where better options than CP do not exist. While the multi-session MLSS test is indeed too cumbersome to be of practical value, there are several single-session MLSS/MMSS alternatives. Although none of the alternatives is physically easy, none requires the absolute maximal efforts that are the conceptual cornerstones of CP testing. Tests such as OBLA, IAT, or D_{max} , are empirically-based, estimating MMSS moderately well on a group-mean basis but are less reliable in individual estimates. Still, they probably all approximate or exceed the typical accuracy associated with CP.

While the above-mentioned tests are well-tested but empirically based alternatives of moderate validity, there presently are two physiologically-based, single-session alternatives that facilitate robust MMSS estimation. Both are directly based on or inspired by the Lactate Minimum Test (LMT) proposed by Tegtbur et al. (1993) eight years following MLSS's introduction by Heck et al. (1985). The LMT was an ingeniously conceived first attempt to capture the highest-attainable $[La^-]$ equilibrium in a physiological rather than empirical manner. A graded exercise test followed induced hyper-lactemia and the exercise intensity at which the lowest $[La^-]$ (lactate minimum) subsequently

occurred was defined as the MLSS estimate. Unfortunately, it eventually became clear that LMT underestimated MLSS because the identified lactate minimum could occur anywhere within the transition zone between the lactate threshold and MLSS intensities (Dotan et al. 2011; Dotan 2012).

Thus, LMT is not recommended for use on its own, but its two subsequent descendants are presented and endorsed below.

The modified LMT

LMT's shortcoming prompted a suggestion of empirical correction to the otherwise physiological-based test (Dotan et al. 2011), but its practicality has not been tested. Wahl et al. (2018), however, developed a different LMT modification, based on adjusting stage durations, which in 18 runners yielded the same mean running velocity as MLSS and a 0.964 ICC correlation between the two tests.

The Reverse Lactate Threshold test

Developed to address both MLSS and LMT's shortcomings, the RLT (Dotan 2012) improved on LMT's under-estimation of MLSS by approaching the threshold from higher rather than lower $[La^-]$ s. This avoided LMT's highest-equilibrium ambiguity and solved MLSS's impracticality by providing a single-session protocol. MLSS's bracketed characteristic and attendant inaccuracy was solved by producing a continuous curve whose apex determines MMSS. The RLT has shown exceptionally high test-re-test reliability ($ICC = 0.997$) in 10 trained and untrained cyclists with a mere 0.75-W (0.3%) mean difference between the two iterations. RLT's validity as MLSS substitutes (as well as that of Wahl's modified LMT), has been demonstrated by Wahl et al. (2017). RLT's validity of predicting running performance has recently been shown by the same group (Wahl et al. 2021).

Summary

1. While it is still impossible to determine whether true CP follows a mathematically precise hyperbolic pattern, or only an approximation thereof, the CP concept remains highly attractive, with its apparent hyperbolic nature making it aesthetically elegant, as well.
2. Although CP appears conceptually valid in determining the highest sustainable exercise intensity, there is overwhelming evidence, beyond possible experimental error and biological variability, that observed CP is unsustainable and significantly higher than MLSS/MMSS intensity.

3. With nearly no exceptions, it can be shown that shorter, higher-intensity CP-determination trials yield higher CP estimates than do longer, less intense ones.
4. Shorter trials produce power outputs that are lower than those predicted by longer trials, or the theoretical hyperbolic power–time curve, based on truly linear work–time and power–1/time relationships. This depresses the higher-intensity segment of the power–time curve and, in a counterclockwise fashion, consequently raises the longer-duration end of the curve, thus elevating the power asymptote and the derived CP.
5. The main reasons for the observed depression of higher-intensity power outputs appear to be energy losses associated with the reciprocating limb masses at progressively higher cadences, increasing muscle activation cost of increasing contractile frequency, increased agonist–antagonist co-contraction, non-optimal cadence, and the decreasing efficiency of increasing loads.
6. The root causes of CP's overestimation of MLSS/MMSS cannot be comprehensively addressed. On an empirical basis, this is partly corrigible via the use of sufficiently long trials and protocol standardization. Available, however, are several empirically based and, in particular, two physiologically founded alternatives to the cumbersome MLSS test. Unlike CP, none of them requires absolute maximal performance, while all of which can provide comparable or better estimates of the true heavy–severe exercise boundary than can CP.

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Declarations

Conflict of interest The author declares no conflict of interest.

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